ORDINANCES
AND
OUTLINES OF TESTS,
SYLLABI AND COURSES OF READING
FOR
BACHELOR OF COMPUTER APPLICATION
BCA Part III (Semester V & VI)
2024-2025 & 2025-2026 SESSIONS

PROGRAMME CODE: BCA

General Shivdev Singh Diwan Gurbachan Singh

KHALSA COLLEGE PATIALA

An Autonomous College
NAAC Accredited 'A' Grade
College with Potential for Excellence Status by UGC,
Star College Status-DBT
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APPROVED

Chairperson
Board of Studies

Preamble:

General Shivdev Singh Diwan Gurbachan Singh Khalsa College Patiala, accredited 'A' grade by NAAC (2015), recognized as "College with Potential for Excellence" status by UGC, New Delhi (2016) and an Autonomous College (2016), is a premier institute of higher education in the state of Punjab since 1960. Being concordant with the need to the creation of a self-sustaining, global knowledge society, the college has undertaken several measures initiated by UGC to bring equity, efficiency and excellence in the Higher Education System of the country.

The important measures taken to enhance academic standards and quality in higher education include innovation and improvements in curriculum, teaching-learning process, examination and evaluation system, besides governance and other matters.

The UGC has formulated various regulations and guidelines from time to time to improve the higher education system and maintain minimum standards and quality across the Higher Educational Institutions in India. The academic reforms recommended by the UGC in the recent past have led to overall improvement in the higher education system. However, due to lot of diversity in the system of higher education, there are multiple approaches followed by Higher Educational Institutions towards examination, evaluation and grading system. While the Higher Educational Institutions must have the flexibility and freedom in designing the examination and evaluation methods that best fits the curriculum, syllabi and teaching–learning methods, there is a need to devise a sensible system for awarding the grades based on the performance of students. Presently, the performance of the students is reported using the conventional system of marks secured in the examinations or grades or both. Then there is conversion from marks to letter grades as the letter grades are used widely across the Higher Educational Institutions in the country. This creates difficulty for the academia and the employers to understand and infer the performance of the students graduating from different universities and colleges based on grades.

The grading system is considered to be better than the conventional marks system and hence, it has been followed in the top institutions in India and abroad. So, it is desirable to introduce uniform grading system. This will facilitate student mobility across institutions within and across countries and will also enable potential employers to assess the performance of students. To bring in the desired uniformity in grading system and method for computing the cumulative grade point average (CGPA) based on the performance of students in the examinations, the UGC has formulated CBCS guidelines.

Outlines of Choice Based Credit System:

1. **Core Course:** A course, which should compulsorily be studied by a candidate as a core requirement is termed as a Core course.
2. **Elective Course:** Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline/ subject of study or which provides an extended scope or which enables an exposure to some other discipline/subject/domain or nurtures the candidate's proficiency/skill is called an Elective Course.

2.1 Discipline Specific Elective (DSE) Course: Elective courses offered under the main discipline/subject of study is referred to as Discipline Specific Elective.

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2.2 Dissertation/Project: An elective course designed to acquire special/advanced knowledge, such as supplement study/support study to a project work, and a candidate studies such a course on his own with an advisory support by a teacher/faculty member is called dissertation/project.

2.3 Generic Elective (GE) Course: An elective course chosen from an unrelated discipline/subject, with an intention to seek exposure beyond discipline/s of choice is called a Generic Elective. The purpose of this category of papers is to offer the students the option to explore disciplines of interest beyond the choices they make in Core and Discipline Specific Elective papers.

P.S.: A core course offered in a discipline/subject may be treated as an elective by other discipline/subject and vice versa and such electives may also be referred to as Generic Elective.

3. Ability Enhancement Courses (AEC): The Ability Enhancement (AE) Courses may be of two kinds: Ability Enhancement Compulsory Courses (AECC) and Skill Enhancement Courses (SEC). "AECC" courses are the courses based upon the content that leads to Knowledge enhancement; i. Environmental Science and, ii. English/Hindi/Modern Indian Language (MIL) Communication. These are mandatory for all disciplines. SEC courses are value-base and/or skill-based and are aimed at providing hands-on-training, competencies, skills, etc.

3.1 Ability Enhancement Compulsory Courses (AECC): Environmental Science, English Communication/Hindi Communication/MIL Communication.

3.2 Skill Enhancement Courses (SEC): These courses may be chosen from a pool of courses designed to provide value-based and/or skill-based knowledge.

4. Practical/tutorials: The practicals/tutorials will be conducted keeping in view the spirit of UGC guidelines as per the needs and requirements of the concerned subject.

Project work/Dissertation is considered as a special course involving application of knowledge in solving / analyzing /exploring a real life situation / difficult problem. A Project/Dissertation work would be of 6 credits. A Project/Dissertation work may be given in lieu of a discipline specific elective paper.

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DEFINITIONS

a. **Academic Year:** Two consecutive (one odd + one even) semesters constitute one academic year.

b. **Course:** Usually referred to, as 'papers' is a component of a programme. All courses need not carry the same weight. The courses should define learning objectives and learning outcomes. A course may be designed to comprise lectures/tutorials/laboratory work/field work/outreach activities/ project work/vocational training/viva/seminars/term papers /assignments/ presentations/self study etc. or a combination of some of these.

c. **Credit Based Semester System (CBSS):** Under the CBSS, the requirement for awarding a degree or diploma or certificate is prescribed in terms of number of credits to be completed by the students.

d. **Credit Point (CP):** The numerical value obtained by multiplying the grade point (GP) by the no. of credit(C) of the respective course i.e. CP= GP X C.

e. **Credit(C):** A unit by which the course work is measured. It determines the number of hours of instructions required per week. One credit is equivalent to one hour of teaching (lecture or tutorial) or two hours of practical work/field work per week, i.e. a course with assigned L-T-P: 3-0-2 or 3-1-0 will be equivalent to 4 credits weight-age course.

f. **Cumulative Grade Point Average (CGPA):** It is a measure of overall cumulative performance of a student over all semesters. The CGPA is the ratio of total credit points secured by a student in various courses in all semesters and the sum of the total credits of all courses in all the semesters. It is expressed up to two decimal places.

g. **Grade Point (GP):** It is a numerical weight allotted to each letter grade on a 10 point scale.

h. **Letter Grade:** It is an index of the performance of students in a said course. Grades are denoted by letters O, A+, A, B+, B, C, P and F.

i. **Programme:** An educational programme leading to award of a degree, diploma or certificate.

j. **Semester Grade point Average (SGPA):** It is a measure of performance of work done in a semester. It is ratio of total credit points (CPs) secured by a student in various courses registered in a semester and the total course credits taken during that semester. It shall be expressed upto two decimal places.

k. **Semester:** Each semester will consist of 15-18 weeks of academic work equivalent to 90 actual teaching days. The odd semester may be scheduled from July to December and even semester from January to June.

l. **Transcript or Grade Card (GC) or Certificate:** Based on the grades earned, a grade certificate shall be issued to all the registered students after every semester. The grade certificate will display the course details (code, title, no. of credits, grades secured) along with SGPA of that semester and CGPA earned till date semester.

m. **Semester Examinations:** The comprehensive examinations conducted for summative evaluation of course. The duration of these examinations shall be 3 and 4 hours for theory and practical courses respectively; and the weight shall be as per the ordinance of relevant programme.

n. **L-T-P:** The prescribed hours/week during a semester for Lecture-Tutorial-Practical to a particular course, in accordance with curriculum prescriptions based on respective nature.

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ORDINANCES FOR BACHELOR OF COMUTER APPLICATIONS

Applicability of Ordinances for the time being in force.

Notwithstanding the integrated nature of a course spread over more than one academic year, the Ordinances in force at the time a student joins a course shall hold good only for the examination held during or at the end of the academic year. Nothing in these Ordinances shall be deemed to debar the College from amending the ordinances subsequently and the amended ordinances, if any, shall apply to all the students whether old or new.

1. BCA is an integrated course comprising of three parts spread over three years. Each part will consist of two semesters. The course of study of BCA shall be divided in six semesters and the final examination will be held at the end of every semester in the months of November/December (for semester I,III &V) and April/May (for semester II, IV & VI) or as fixed by the College.

2. The examination in BCA Part-I shall be open to a student who produces the following certificates to the Principal of the college.

(i) of having passed at least +2 examination of Punjab School Education Board / C.B.S.E. / I.C.S.E. or any other examination recognized as equivalent.

OR

of having passed three year Diploma from any recognized technical state education Board.

Note: Candidate placed under reappear in one or more subjects in 10+2 examination of Punjab School Education Board or any other examination, recognized as equivalent thereto shall not be eligible for admission to BCA Part-I Course.

(ii) of having remained on the rolls of a School/college admitted to the privileges of the College for the academic year preceding the examinations.

(iii) of having good character.

(iv) A student will produce proof for opting Elementary Punjabi. This can be opted only under the following conditions:

(a) Those students who have passed their Matric Examination outside the State of Punjab and have not opted for Punjabi Subject.

(b) Wards of Defense Personnel/Para-Military Personnel can opt for Elementary Punjabi.

(c) Children of NRI, NRE and Foreign Students.

2.1 To qualify for admission to 3rd semester of the course, the candidate must have passed 50% of total papers of the two semesters of the 1st year. In case, the result of 2nd semester is not declared at the time of admission to 3rd semester, the student may be admitted provisionally and will be allowed to take examination of 3rd semester if he/she has passed in 50% of the total papers of first year (i.e. 1st and 2nd semester). Similarly, to qualify for admission to 5th semester of the course, the student may be admitted provisionally if the result of the previous semester has not been declared and will be allowed to take examination of 5th semester, if he/she has passed 50% of the total papers of previous semesters.

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2.2 A student who joins BCA Part II course after having passed BCA Part I or any other Equivalent examination from any other statutory University shall be required to qualify all the deficient papers as provided in the syllabus prescribed by the College for BCA Part I examination along with Part II examination.

A student who joins BCA Part III course after having passed BCA Part II or any other equivalent examination from any other statutory University shall be required to qualify all the deficient paper as provided in the syllabus prescribed by this college for BCA Part I and Part II examination along with Part III examination. However, Mid Semester migration will not be allowed.

3 A candidate must complete and pass the whole course of three years within a maximum of six years from the date of admission in BCA First semester. If the candidate does not clear the lower examination within stipulated period, the higher result of the candidate will stand automatically cancelled.

4 Semester examination will be open to regular candidates who have been on the rolls of the college and meet the attendance and other requirements as prescribed in the ordinances of the course.

5 Examination Rules

5.1 Paper Setting/Evaluation will be done by an External Examiner or as decided by the Examination Cell.

5.2 Viva Voce/ Practical examination shall be conducted by a Committee consisting of the following:

(i) One external examiner

(ii) One internal examiner

5.3 The supplementary examination will be held along with the routine End Semester Tests. The supplementary paper would be from the syllabi prescribed for that session in which the candidate is appearing. The student can appear in the theory/practical paper on the payment of the required fee. The candidate will have consecutive two attempts to clear the Supplementary Examination, marks of practical and internal assessment will be carried forward as original.

5.4 Re-evaluation of answer sheet in two subjects is allowed after paying the requisite fee. The application for Re-evaluation should be submitted within 15 days of the declaration of the results. In case there is a difference of more than 10% between the marking of the first evaluator and the second evaluator, then the paper would be sent to a third evaluator. The mean of the marks of the second and third evaluators is then considered as the final marks. The re-evaluated marks will be considered final irrespective of the increase or decrease in marks.

5.5 The students who have reappear in the Vth semester only of three year degree course at undergraduate level, will be allowed to appear in their Reappear examination along with the Final Semester Examinations of their respective courses.

5.6 The Principal can provide Golden Chance (with special chance fee) to students who have been unable to clear their exams even after two attempts.

5.7 IMPROVEMENT EXAMINATIONS:

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I. A student who has been declared 'pass' in the Undergraduate course he/she was admitted to, may apply for improvement within a year from the declaration of the result of the final semester and he/she can take maximum of 50% of the total papers for that course for improvement.

II. A student shall have to appear in End semester examination of the paper chosen for improvement along with the regular students. No special exam shall be held for him/her.

III. If a student fails to improve upon the original marks obtained in the paper chosen for improvement, his/her original marks shall be retained and he/she shall not get a second chance for improvement.

IV. Improvement examination in practical/MST paper shall not be allowed.

V. A student taking improvement examination shall have to pay a fee decided by the college.

5.8 Grading System:

The grades and their description, along with equivalent numerical grade points are listed in the Grading Assignment Table below:

Grade Assignment Table

Percentage of Marks	Description	Grade	Grade Point
91-100	Outstanding	O	10
81-90	Excellent	A+	9
71-80	Very Good	A	8
61-70	Good	B+	7
51-60	Above Average	B	6
41-50	Average	C	5
35-40	Pass/Fair	P	4
0-34	Fail	F	0
Otherwise	Absent/Detained	Ab/D	0

a. A student obtaining Grade F shall be considered failed and will be required to reappear in the examination.

b. For non credit courses **'Satisfactory' or 'Unsatisfactory'** shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.

5.9 Computation of SGPA and CGPA

The UGC recommends the following procedure to compute the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA):

a. The SGPA is the ratio of sum of the product of the number of credits with the grade points scored by a student in all the courses taken by a student and the sum of the number of credits of all the courses undergone by a student, i.e.
SGPA (Si) = ∑(Earned Credits Ci x Grade Point Gi)/∑Earned Credits Ci;
Where Ci is the number of credits of the ith course and Gi is the Grade Point Scored by the student in the ith course.

b. The CGPA is also calculated in the same manner taking into account all the courses undergone by a student over all the semesters of a programme, i.e.
CGPA (Ci) = ∑(Earned Credits Ci x SGPA Si)/∑Ci);

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Where Si is the SGPA of the ith semesters and Ci is the total number of credits in that semester.

c. The SGPA and CGPA shall be rounded off to 2 decimal points and reported in the transcripts.

Illustration of the computation of SGPA and CGPA and Format for Transcripts

i. Computation of SGPA and CGPA

Illustration for SGPA

Course	Credits	Grade Letter	Grade Point	Credit Point (Credit x Grade)
Course 1	3	A	8	3 X 8 = 24
Course 2	4	B+	7	4 X 7 = 28
Course 3	3	B	6	3 X 6 = 18
Course 4	3	O	10	3 X 10 = 30
Course 5	3	C	5	3 X 5 = 15
Course 6	4	B	6	4 X 6 = 24
	20			139

Thus, **SGPA = 139/20 = 6.95**

Illustration for CGPA

Semester 1	**Semester 2**	**Semester 3**	**Semester 4**
Credit : 20 SGPA : 6.9	Credit : 22 SGPA : 7.8	Credit : 25 SGPA : 5.6	Credit : 26 SGPA : 60
Semester 5	**Semester 6**		
Credit : 26 SGPA : 6.3	Credit : 25 SGPA : 8.0		

Thus, **CGPA** = $\frac{20 \times 6.9 + 22 \times 7.8 + 25 \times 5.6 + 26 \times 6.0 + 26 \times 6.3 + 25 \times 8.0}{144} = 6.73$

ii. Transcripts (Format):

Based on the above recommendations on Letter grades, grade points, SGPA and CGPA, the College may issue the transcript for each semester and a consolidated transcript indicating the performance in all semesters.

5.10 Division and Position:

Division shall be awarded in the following manner, to the candidates on the basis of their respective CGPA:

CGPA 7.5 or more	1st Division with Distinction
CGPA 6.0 or more but less than 7.5	1st Division
CGPA 5.0 or more but less than 6.0	2nd Division
CGPA 3.5 or more but less than 5.0	3rd Division
Otherwise	Fail

However, First, Second or Third position shall be awarded to the candidates, provided they meet the following conditions:

a) Rank shall be solely decided on the final CGPA, on completion of degree credit requirement.

b) The candidate has completed all the prescribed requirements, in the prescribed programme duration.

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c) The candidate has passed / secured valid grades in all the prescribed courses, in the first attempt.

d) No disciplinary action is pending or has ever been lodged against him/her.

e) In case of an exceptional tie, both candidates shall be awarded the same rank.

5.11 Grade Card:

At the end of each semester, a student will be given a 'Grade Card' which will contain Course Code, Title, Credits, Grades Awarded, Earned Credits and Earned Point secured by him/her in each course, together with his/her SGPA in that semester. On the completion of the programme, a Final Grade Card will be issued to the student, giving full semester-wise details about the absolute marks and grades obtained by him/her in each course together with his/her SGPA and also the CGPA and Division awarded to him/her.

5.12 Equivalence:

Percentage (P) equivalent to CGPA earned by a candidate may be calculated using the following formula:

$$P = CGPA \times 10$$

5.13 MALPRACTICES/UNFAIR MEANS

5.13.1 The following shall be deemed to be unfair means:

I. Leaving the Examination Hall without submitting the answer book to the invigilator or taking away, tearing off or otherwise disposing off the same or any part thereof.

II. Using abusive language in the examination hall or writing the same in the answer sheet.

III. Making an appeal to the evaluator through answer sheet.

IV. Possession by examinee or having access to books, notes, papers, mobile or any other electronic material which can prove to be helpful in the exam.

V. Any action on the part of candidate at an examination trying to get undue advantage in the performance at examinations or trying to help another, or derive the same.

VI. Case of Impersonation in the examination.

VII. Intimidating, threatening, manhandling, using violence, show of force in any form against any invigilator or any person on duty, creating disturbance to the smooth conduct of the examination.

VIII. Any other action which the Controller Examination / Chief Controller deem fit to be a case of UMC.

5.13.2 In case the student is found to have used any of the above Unfair means:

I. His/her answer book shall be seized and He/She will be given a new answer sheet.

II. Invigilator shall submit a detailed report along with the answer book of the student and the related material, if any, to the Centre Superintendent who will subsequently hand it over to Controller Examination.

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III. Written statement to this effect shall be obtained from the student by the Centre Superintendent. In case the student refuses to do the same, the fact of refusal must be recorded.

IV. The student reported to have used unfair means shall be allowed to appear in the subsequent papers. However, no marks would be awarded for the paper in which unfair means were used.

V. The Principal shall refer the cases of malpractices in Mid Semester tests, House Tests and End Semester Examinations, to an Unfair Means Committee, constituted by him/her for the purpose. Such committee shall follow the approved scales of punishment. The Principal shall take necessary action, against the erring students based on the recommendations of the committee.

5.13.3 The involvement of the Staff, who are in charge of conducting examinations, evaluating examination papers and preparing/keeping records of documents relating to the examinations if involved in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and recommended for award of appropriate punishment after enquiry.

6 Attendance Regulations & Condonation:

6.1 A student shall be eligible to appear for end semester examinations, if he/she acquires a minimum of 75% of attendance in each subject.

6.2 Request to the Principal for Condonation of shortage of attendance after the recommendation of the HOD will be forwarded to Lecture Shortage Condonation Committee. The committee can finally condone the shortage in aggregate up to 15% on medical grounds in each semester.

6.3 Any student representing the Institute/ University/ State/ Nation in any Academic/ Sports/ Cultural/Extra Co curricular/ NSS/NCC or any other event shall be considered on duty. His/ Her shortage of lectures shall be condoned, provided that the student is permitted in writing by the Principal/HOD concerned and a certificate to this effect signed by the competent authority where the student attended the event is taken.

6.4 A Student will not be promoted to the next semester unless he/she satisfies the attendance requirement of the present semester as applicable.

6.5 Students whose shortage of attendance is not condoned in any semester are not eligible to take their end semester examination of that particular semester and their registration for examination shall stand cancelled and no fee shall be refunded.

7. Late college students: A candidate, who has completed the prescribed course of instructions for a semester but has not appeared in the examination or having appeared, has failed in the examination, may appear as a late college student within the prescribed period.

8. Applications for admission to the examination shall be made on the prescribed form attested by the competent authority as per the college rules.

9. Amount of examination fee to be paid by a candidate for each semester shall be as fixed by the College from time to time.

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10. The last date by which examination forms and fees must reach the College Office shall be as follows.

Semester	Without late fee	With late fee of Rs. 800/-	With late fee of Rs.1200/-	With late fee of Rs.5000/-	With late Fee of Rs. 10,000
Nov./Dec. (Odd)	Sept. 30	Oct.15	Oct. 21	Oct. 31	Nov. 10*
April/May(Even)	Feb. 28	March 15	March 21	March 31	April 15*

***Note: No Examination Form will be accepted after the prescribed date.**

11. College medal will be awarded to a candidate who secures first position in the College on the basis of the marks of all the six semesters taken together. The general rules and conditions of the College/University for the Award of medal/prizes etc. will be applicable in the award of College medal to the topper of this examination.

12. All the question papers except Punjabi Language Paper will be in English and candidates can answer the questions only in English.

13. The minimum number of marks required to pass each semester examination will be 35% in each paper and 35% in the aggregate of the semester examination. Provided, that in papers with practicals, the percentage shall be required separately in written and practical/lab work.

14. Assessment:.

14.1 The Assessment in each semester of BCA Course will be 30% internal and 70% external for each Theory paper. The result of the Internal Assessment shall be conveyed to the students/examination branch by the Head of the Department according to prescribed schedule.

14.2 There shall be Two Mid Semester tests in each Semester.

14.3 Internal Assessment of 30% will be based on Continuous Comprehensive Assessment (CCA) pattern and the breakup of 30% will be as under:

(i) Averege of Two mid Semester Tests : 40%

(ii) Assignment/Seminar/Class Test/Tutorial/Quiz/Internal Practical etc. : 40%

(iii) Attendance : 20%

Papers having practical/viva, the marks of theory and practical/viva will be reduced equally percentage wise to make room for 30% internal assessment.

Note: If a case comes to notice of Controller of Examinations where the marks awarded by the Teacher are on a very Higher/Lower side, the award will be got moderated by the following committee.

I. Paper Evaluator
II. Head of the Department
III. Controller of Examination

14.4 A candidate is required to secure at least 35% marks both in external examination (Theory and Practical/ Project work) and in internal assessment separately in each paper in order to qualify in an examination.

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14.5 In case the student is dissatisfied with the marks awarded to him/her in internal assessment; he/she can approach the concerned teacher. If the student is still not satisfied, he/she may approach the head of department and the Principal subsequently.

15. End-Semester Examination:

End-semester examination(s) of each theory course shall be of three hours duration and will be conducted as per norms and schedule notified by the Controller of Examination. The end semester examinations of laboratory/practical courses and other courses such as seminar, colloquium, field work, project, dissertation etc. shall be conducted as notified by the HOD.

16. Degree Requirement:

16.1 An undergraduate degree with Honours in a discipline may be awarded if a student completes atleast 14 core papers in that discipline, 2 Ability Enhancement Compulsory Courses (AECC), minimum 2 Skill Enhancement Courses (SEC)and 4 papers each from a list of Discipline Specific Elective and Generic Elective papers, respectively.

An undergraduate degree in Science disciplines may be awarded if a student completes 14 core papers each in three disciplines of choice, 2Ability Enhancement Compulsory Courses (AECC), minimum 4 Skill Enhancement Courses (SEC)and 2 papers each from a list of Discipline Specific Elective papers based on three disciplines of choice selected above, respectively.

An Undergraduate degree in Humanities/ Social Sciences/ Commerce may be awarded if a student completes atleast 4 core papers each in two disciplines of choice, 2 core papers each in English and Hindi/MIL, respectively, 2 Ability Enhancement Compulsory Courses (AECC), minimum 4 Skill Enhancement Courses (SEC), 2 papers each from a list of Discipline Specific Elective papers based on the two disciplines of choice selected above, respectively, and two papers from the list of Generic Electives papers.

16.2 The result of all the examinations will be declared through the College website.

16.3 The grace marks shall be allowed according to the general ordinances relating to 'Award of Grace Marks'. These ordinances will apply to all examinations.

(i) Grace marks to be given shall be calculated on the basis of 1% of total aggregate marks of all the written and practical papers of the examination concerned. Marks for viva-voce/internal assessment/sessional work/skill in teaching/any additional /optional /deficient subject shall not be taken into account for this purpose. If a fraction works out to half or more, it shall count as one mark and fractional less than half shall be ignored

(ii) To pass in one or more written papers or subjects, and/or to make up the aggregate to pass the examination but not in sessional work, internal assessment, viva-voce and skill in teaching.

16.4 The College may from time to time revise, amend and change the regulations or the curriculum, if found necessary.

16.5 The students will be given the facilities of transfer of Credits earned in different recognized/approved Institutions of Higher Education in India and Abroad.

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16.6 A student who earns total specified credits according to the curriculum and fulfills such other conditions as may be mentioned in the curriculum of the programme, shall be issued the DMC and shall be awarded degree by Punjabi University Patiala. He/she must also pay all College dues as per rules. Moreover, there should be no case of indiscipline pending against him/her.

17. If any student gets admission after concealing any fact or his/her certificates are found fake after verification or he/she misleads the institution as any front or because of any other reason, his/her admission will stand cancelled/ his/her result cancelled though he/she may have been declared pass.

18. In case the ordinance is silent about any issue, it will be decided by the College Principal in consultation with the Academic Advisory Committee of the college in the anticipation of approval of the same by Academic Counsel of the College.

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BACHELOR OF COMPUTER APPLICATION (BCA) III
Session- 2024-2025 & 2025-2026
COURSE STRUCTURE UNDER LOCF BASED ON CBCS

Course Code	Course Title	Course Type	Lectures	Tutotials	Practicals	Credits	Marks
		SEMESTER V					
BCA 501	Java Programming	Core Course C- 9	**3**	**1**		**4**	**100**
BCA 501(P)	Software Lab-VIII (Based on BCA 501)				2	**2**	**50**
BCA 502	Web Technology-I	Core Course C-10	**3**	**1**		**4**	**100**
BCA 502(P)	Software Lab-IX (Based on BCA 502)				2	**2**	**50**
BCA 503	Software Engineering	Discipline-Specific Elective(DSE)-1	**3**	**1**		**4**	**100**
BCA 504	Punjabi Compulsory	MIL-5	**5**		1	**4**	**100**
BCA 504 A	Punjabi Compulsory (Mudla Gyan/Elementary Punjabi)						
BCA 505	English Communication Skills	Ability – Enhancement Compulsory Course(AECC)-7	**3**		1	**4**	**100**
BCA 506	**Any one of the following** (A) Management Information System (B) Artificial Intelligence	Discipline-Specific Elective(DSE)-2	**3**		1	**4**	**100**
	Gross Total Credit Point					**28**	**700**

SEMESTER VI

Course Code	Course Title	Course Type	Lectures	Tutotials	Practicals	Credits	Marks
BCA 601	Web Technology – II	Core Course C-11	**3**	**1**		**4**	**100**
BCA 601(P)	Software Lab - X (Based on BCA 601)				2	**2**	**50**
BCA 602	E- Technologies	Discipline-Specific Elective(DSE)-3	**3**	**1**		**4**	**100**
BCA 603	Minor Project	Skill Enhancement Elective Course (SEC)-3			4	**2**	**50**
BCA 604	System Software	Discipline-Specific Elective(DSE)-4	**3**	**1**		**4**	**100**
BCA 605	Punjabi Compulsory	MIL-6	**5**		1	**4**	**100**
BCA 605 A	Punjabi Compulsory (Mudla Gyan /Elementary Punjabi)						
BCA 606	English Communication Skills	Ability – Enhancement Compulsory Course(AECC)-8	**3**		1	**4**	**100**
BCA 607	**Any one of the following**	Skill Enhancement Elective Course(SEC)-4	**3**	**1**		**4**	**100**
	BCA 607 (A) Cloud Computing						
	BCA 607 (B) Digital Marketing						
	Gross Total Credit Point					**28**	**700**

SEMESTER V
BCA 501: JAVA PROGRAMMING
4 Credits: (3L+1T)

Time: 3 hours **Maximum Marks: 100 Marks**
Pass Marks: 35% **Internal Assessment: 30 Marks**
Theory: 70 Marks

Course Objective

The aim of this course is to enable students to develop robust and faster object-oriented applications using the concepts of multithreading and exception handling.

Course Learning Outcomes

After completion of this course the students will be able to:

CO1: Implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity.

CO2: Identify classes, objects, members of a class and the relationships among them needed for a finding the solution to specific problem.

CO3: Demonstrate how to achieve reusability using inheritance, interfaces and packages and describes faster application development can be achieved.

CO4: Demonstrate understanding and use of different exception handling mechanisms and concept of multithreading for robust faster and efficient application development.

INSTRUCTIONS FOR THE PAPER SETTER

The question paper will consist of three sections A, B & C. Section A and B will have four questions each from respective unit of the syllabus carrying 10 marks for each question. Section C will have 10-15 short answer type questions, carrying total 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES

The candidates are required to attempt at least two questions each from sections A and B, and the entire section C.

UNIT-I

Introduction to Java: Evolution, Features, Difference between C++ and Java, Java Virtual Machine and Byte code.

Classes, Objects and Methods: Class Specification, Class Header and Body, Declaring Objects, Defining Methods, Method Overloading, Passing Arguments, Returning Objects, Static Fields and Static Methods, Nested and Inner Classes.

Constructor: Defining Constructor, Parameterized Constructor, Default Constructor, Copy Constructor, use of this keyword, Garbage Collection, Arrays, Strings.

Inheritance: Types of Inheritance, Use of super keyword, Method Overriding, Abstract Methods and Classes, Final Methods and Classes, Defining an Interface, Implementing Interface, Packages.

UNIT-II

Exception handling: Difference between Errors and Exceptions, Types of Exception, try catch and final block, throw, throws and built in Exceptions.

Multithreaded Programming: Creating Threads, Life cycle of Thread, Thread priority, Thread, Synchronization and Thread Communication.

Event Handling: Event Classes, Event Sources, Event Delegation Model , Event Listener Interfaces, Adapter Classes.

Swing: Features, Swing Packages, Components and containers, working with Swings, User Interface Components: JApplet, Label, Button, Checkbox, Text Field, Text Area, Radio Button, Panel, Scroll Pane and List. Types of Layouts: Flow Layout, Border Layout, Grid Layout, Card Layout. **JDBC:** JDBC Fundamentals, Creating tables in Database, Establishing Connectivity.

Suggested Readings:

1. Patrick Naughtonand Herbert Schildt, "The Complete Reference Java 2", Tata McGraw Hill.
2. L Lemay, Teach yourself Java in 21 days", Sams Publishers.
3. Mark C. Chan,"1001 Java Programming Tips", Jamsa Press.
4. E Balagurusamy, "Programming with Java", Tata McGraw Hill.

Approved

Chairperson
Board of Studies

BCA 501 (P): SOFTWARE LAB VIII (BASED ON PAPER BCA 501)
2 Credits: 4H (P)

Maximum Marks: 50 **Maximum Time: 3 hours**
Internal Assessment: 15 Marks **Pass Marks: 35%**

Course Objective

The aim of this course is to enable students to develop robust and faster object-oriented applications using the concepts of multithreading and exception handling.

Course Learning Outcomes

After completion of this course the students will be able to:

CO1: Implement Object Oriented programming concept of control Structures, strings and function for developing the logic building activity using Java.

CO2: Identify classes, objects, members of a class and the relationships among them needed for finding the solution of problems by designing the programs.

CO3: Implement programs related to reusability using inheritance, interfaces and packages.

CO4: Design and implement programs of exception handling mechanisms and multithreading for efficient application development.

This laboratory course will comprise as exercises to supplement what is learnt under paper BCA 501-Java Programming.

The breakup of marks for the practical will be as under:-

i. Internal Assessment 15Marks
ii. Viva Voce(External Evaluation) 15Marks
iii. Lab Record, Program Development and Execution(External Evaluation) 20Marks

Students are required to develop the following programs with internal documentation to demonstrate:

1. Concept of class and object in java.
2. Method overloading.
3. Static fields and static methods
4. Nested classes.
5. Copy constructor.
6. Constructor overloading.
7. Concept of this keyword.
8. Matrix addition.
9. Five string operations.
10. Single inheritance using super keyword.
11. Multilevel inheritance.
12. Method overriding.
13. Implements interface.
14. Importing multiple packages.
15. Use of exception handling and use of finally keyword.
16. Create threads by extending Thread class.
17. Create threads by implementing Runable interface.
18. Demonstrate event handling mechanism using keyListener interface.
19. Create swing application.
20. Implement flow layout.
21. Implement JDBC.

Approved

Chairperson
Board of Studies

BCA 502: WEB TECHNOLOGY-I

4 Credits: (3L+1T)

Time: 3hours **Maximum Marks: 100 Marks**
Pass Marks: 35% **Internal Assessment: 30 Marks**
Theory: 70 Marks

Course Objective

The objective of this course is to help the students to acquire the basic web development skills using HTML, CSS, JavaScript and AJAX.

Course Learning Outcomes

After completion of this course students will be able to:

CO1: Understand the major areas and the challenges of latest technologies of internet.
CO2: Enhance the interface between the various languages.
CO3: Explore the various ideas for database access and data driven applications.
CO4: Develop the skills in analyzing the usability of the websites.

INSTRUCTIONS FOR PAPER-SETTER

The question paper will consist of three sections A, B and C. Section A and B will have four questions from the respective section of the syllabus carrying 10 marks for each question. Section C will consist of 10-15 short answer type questions carrying a total 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES

Candidates are required to attempt at least two questions each from sections A and B and the entire section C.

UNIT - I

HTML and Overview: Basic HTML concepts, an overview of HTML5 markup, the phases of Web site development, Structure of HTML document.

HTML and Images: The role of images on the Web, image preliminaries, Obtaining images; HTML image basics, Image as buttons, Image maps.

Layout: Backgrounds, Colors, and Text**,** Design requirements; HTML approach to Web Design, Fonts, Colors in HTML.

Tables: Create a table, Row span, Col span, Attributes of table tag, Add image background in table. Introduction to List, Types of list. Introduction to frames.

UNIT - II

Forms: The <FORM> elements, form controls.

CSS: Concept of CSS, Creating Style Sheet, CSS Properties, CSS Styling (Background, Text Format, Controlling Fonts), CSS Id and Class, Introduction to Bootstrap and tailwind.

Java script: Introduction, documents, statements, functions, objects, form validation.

AJAX: Introduction to JQuery; Definition, working and components of AJAX, The XML Http Request object, callback function.

Suggested Readings:

1. Thomas A. Powe**l**, “HTML: The Complete Reference”,Osborne/McGraw-Hill.
2. Deitel, " Deitel and Nieto: Internet &WWW. How to program",Pearson Education Asia.
3. Steven M. Schafer, “HTML, CSS, JavaScript, Perl, Pythonand PHP”, Wiley India.
4. Kogent Learning Solutions Inc, "Photoshop CS6 in Simple Steps", Dreamtech Press.
5. Brad Wi**l**iams, David Damstra, and Hal Stern, “Professional WordPress: Design and Development”,John Wiley &Sons.

Approved

Chairperson
Board of Studies

BCA 502 (P): SOFTWARE LAB-IX (BASED ON PAPER BCA 502)
2 Credits: 4H (P)

Maximum Marks: 50 **Maximum Time: 3 hours**
Internal Assessment: 15 **Marks Pass Marks: 35%**

Course Objective
The objective of this course is to help the students to acquire the basic web development skills using HTML, CSS, JavaScript and AJAX.
Course Learning Outcomes
After completion of this course students will be able to:
CO1: Understand basic elements related to the latest technologies of internet.
CO2: Enhance the interface between various languages for effective Website development. **CO3:** Explore and implement the various ideas for database access and data driven applications. **CO4:** Develop the skills in analyzing the usability of the websites.

This laboratory course will comprise as exercises to supplement what is learnt under paper BCA 502- Web Technology-I.

The breakup of marks for the practical will be as under: -

i. Internal Assessment 15Marks
ii. Viva Voce (External Evaluation) 15Marks
iii. Lab Record, Program Development and Execution (External Evaluation)20 Marks Students are

required to develop the following programs with internal documentation:

1. Introduction to HTML basic tags.
2. Write a program to create lists (ordered, unordered and definition list).
3. Write a program to create tables.
4. Write a program to display images on a webpage.
5. Write a program to create hyperlinks.
6. Write a program to create HTML forms.
7. Write a program to create HTML Frames.
8. Write a program to validate a form using HTML5.
9. Create a webpage using HTML and CSS.
10. Write a program to create a menu using HTML and CSS.
11. Write a program to print date using JavaScript.
12. Write a program to Show use of alert, confirm and prompt box.
13. Create validation Form in JavaScript.
14. Create a responsive webpage using Bootstrap.
15. Create a responsive webpage using tailwind.
16. Write a program to manipulate CSS properties using JQuery.
17. Write a program to change attributes of HTML elements using JQuery.
18. Write a program to load the content of a text file into HTML div element using AJAX.

Approved

Chairperson
Board of Studies

BCA 503: SOFTWARE ENGINEERING
4 Credits: (3L+1T)

Time: 3hours **Maximum Marks: 100**
Pass Marks: 35% **Internal Assessment: 30 Marks**
Theory: 70Marks

Course Objective

This course aims to demonstrate the software development process and to provide better understanding of design, coding and testing phases of project development. It will also help students to understand the importance of documentation in software development.

Course Learning Outcomes

After completion of this course students will be able to:

CO1: Analyse software development process models, including spiral model and traditional models like waterfall.
CO2: Apply software life cycle through requirements gathering, choice of process model and design model for developing projects.
CO3: Acquire knowledge about software estimation techniques and configuration management for creating project baselines.
CO4: Perform various activities like implementing, testing and maintenance.

INSTRUCTIONS FOR THE PAPER SETTER

The question paper will consist of three sections A, B & C. Section A and B will have four questions each from respective unit of the syllabus carrying 10 marks for each question. Section C will have 10 -15 short answer type questions, carrying total 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES

The candidates are required to attempt at least two questions each from sections A and B, and the entire section C.

UNIT-I

Software Engineering – Definition, Goals, The Problem Domain, Software Engineering Challenges, Role of Software Engineering, Software development life cycle, its phases**,** Software development process, models: Waterfall, Prototyping, Iterative, and Spiral.

Planning a Software Project: Cost estimation, Single variable model, and COCOMO model, Software configuration management process.

Software Metrics–Definition, Importance, Categories of metrics. Software Quality–attributes Cyclomatic complexity metric.

Software Requirements Analysis –Requirement Process, Types of Requirement Process, Need for SRS, Data flow diagrams, Data Dictionary, entity relationship diagram, Characteristics and components of SRS.

UNIT-II

Software Design – Design Activities, Design principles, Module-level concepts, Structure Chart and Structured Design methodology.

Coding: Top–down and Bottom–up Approaches, Structured Programming, Programming style, verification code inspection, Static Analysis.

Software Testing – Testing Process and Goals, Black Box Testing: Equivalence class partitioning, Boundary value analysis, cause-effect graphing; White Box Testing: Control flow and Data flow based testing, mutation testing; levels of testing, test plan, test case specification, test case execution and analysis.

Software Maintenance – The Nature and types of Maintenance, Maintenance process and problems, Software re-engineering, Reverse engineering, forward engineering.

Approved

Chairperson
Board of Studies

ਭਾਗ-ਤੀਜਾ, ਸਮੈਸਟਰ-ਪੰਜਵਾਂ

BCA 504- ਪੰਜਾਬੀ ਲਾਜ਼ਮੀ

ਕ੍ਰੈਡਿਟ : 4 **ਅਧਿਆਪਨ ਦੇ ਪੀਰੀਅਡ ਪ੍ਰਤੀ ਹਫ਼ਤਾ : 05**

ਕੁੱਲ ਅੰਕ : 100 **ਵਿਸ਼ੇ ਵਿੱਚੋਂ ਪਾਸ ਅੰਕ : 35%**

ਲਿਖਤੀ ਪ੍ਰੀਖਿਆ : 70 ਅੰਕ **ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ : 30 ਅੰਕ**

ਨੋਟ : ਵਿਦਿਆਰਥੀਆਂ ਵੱਲੋਂ ਪਾਸ ਹੋਣ ਲਈ ਲਿਖਤੀ ਪ੍ਰੀਖਿਆ ਅਤੇ ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ ਦੇ ਅੰਕਾਂ ਵਿਚੋਂ ਅਲੱਗ-ਅਲੱਗ ਤੌਰ 'ਤੇ 35% ਅੰਕ ਲੈਣੇ ਜ਼ਰੂਰੀ ਹਨ

Course Outcome

1.ਜੀਵਨੀ ਤ ਸੇਧ ਪਾਪਤ ਕਰਕੇ ਵਿਦਿਆਰਥੀ ਜਿੰਦਗੀ ਵਿੱਚ ਆਉਣ ਵਾਲੀਆਂ ਔਕੜ ਦੇ ਬਾਵਜੂਦ ਕਠਿਨ ਮਿਹਨਤ ਰਾਹੀਂ ਨਿਰੰਤਰ ਅੱਗੇ ਵਧਣ ਅਤੇ ਆਪਣੇ ਨਿਸ਼ਾਨਿਆਂ ਤੇ ਪਹੁੰਚਣ ਦਾ ਹੁਨਰ ਸਿੱਖਦਾ ਹੈ।

2. ਵਿਦਿਆਰਥੀ ਇਕ ਦੌੜਾਕ ਜ ਖਿਡਾਰੀ ਲਈ ਸਿਹਤ,ਖੇਡ ਅਤੇ ਸਮਾਜ ਸੰਬੰਧੀ ਜਰੂਰੀ ਨੁਕਤਿਆਂ ਬਾਰੇ ਜਾਣਕਾਰੀ ਹਾਸਲ ਕਰਦਾ ਹੈ।

3. ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦੀ ਵਿਆਕਰਣਕ ਇਕਾਈ ਵਾਕੰਸ਼ ਬਾਰੇ ਜਾਣਕਾਰੀ ਹਾਸਲ ਕਰਦਾ ਹੈ ਜਿਹੜੀ ਜਿੱਥੇ ਉਸ ਦੀ ਭਾਸ਼ਾ ਕੁਸ਼ਲਤਾ ਵਿਚ ਵਾਧਾ ਕਰਦੀ ਹੈ ਥੇ ਦੂਜੀਆਂ ਭਾਸ਼ਾਵ ਨੂੰ ਸਿੱਖਣ ਵਿਚ ਵੀ ਸਹਾਈ ਹੁੰਦੀ ਹੈ।

4. ਵਿਦਿਆਰਥੀ ਚਲੰਤ ਮਸਲਿਆਂ ਬਾਰੇ ਜਾਣਕਾਰੀ ਹਾਸਿਲ ਕਰਕੇ ਇਹਨ ਵਿਚ ਸੁਧਾਰ ਕਰਨ ਲਈ ਵਚਨਬੱਧ ਹੁੰਦਾ ਹੈ।

ਪਾਠ-ਕ੍ਰਮ ਅਤੇ ਪਾਠ-ਪੁਸਤਕ

ਭਾਗ- ੳ

ਉਡਣਾ ਸਿੱਖ ਮਿਲਖਾ ਸਿੰਘ (ਜੀਵਨੀ) - ਪ੍ਰਿੰਸੀਪਲ ਸਰਵਨ ਸਿੰਘ

ਭਾਗ- ਅ

I. ਵਿਆਕਰਨ

(ੳ) ਨਾਂਵ ਵਾਕੰਸ਼

(ਅ) ਕਿਰਿਆ ਵਾਕੰਸ਼

II. ਲੇਖ ਰਚਨਾ (ਵਿਦਿਅਕ, ਸਭਿਆਚਾਰਕ ਅਤੇ ਚਲੰਤ ਮਸਲਿਆਂ ਨਾਲ ਸਬੰਧਿਤ)

ਭਾਗ- ੲ

ਭਾਗ ੳ ਅਤੇ ਅ.I ਵਿਅਕਰਨ ਵਾਲੇ ਭਾਗ 'ਤੇ ਆਧਾਰਿਤ ਸੰਖੇਪ ਉਤਰਾਂ ਵਾਲੇ ਪੰਦਰਾਂ ਪ੍ਰਸ਼ਨ।

ਅੰਕ ਵੰਡ ਅਤੇ ਪੇਪਰ ਸੈਟਰ ਲਈ ਹਦਾਇਤਾਂ

1. ਪ੍ਰਸ਼ਨ ਪੱਤਰ ਨੂੰ ਉਪਰੋਕਤ ਨਿਰਧਾਰਿਤ ਪਾਠ ਕ੍ਰਮ ਦੇ ਅਨੁਸਾਰ ਤਿੰਨ ਭਾਗਾਂ (ੳ,ਅ ਅਤੇ ੲ) ਵਿੱਚ ਵੰਡਿਆ ਜਾਵੇ।
2. ਭਾਗ ੳ ਵਿੱਚੋਂ ਜੀਵਨੀ ਦਾ ਸਾਰ/ ਕਲਾ ਪੱਖ/ ਮਿਲਖਾ ਸਿੰਘ ਦੀ ਸਖਸ਼ੀਅਤ ਉਸਾਰੀ (ਦੋ ਵਿੱਚੋਂ ਇਕ) 12 ਅੰਕ
3. ਭਾਗ ੳ ਵਿੱਚੋਂ ਜੀਵਨੀ ਵਿਚਲੀਆਂ ਪ੍ਰਮੁੱਖ ਘਟਨਾਵਾਂ ਸੰਬੰਧੀ ਪ੍ਰਸ਼ਨ (ਚਾਰ ਵਿੱਚੋਂ ਦੋ) 4+4=8 ਅੰਕ
4. ਵਿਆਕਰਨ ਵਾਲੇ ਭਾਗ ਅ-I ਵਿੱਚੋਂ ਵਰਣਾਤਮਕ ਪ੍ਰਸ਼ਨ। (ਦੋ ਵਿੱਚੋਂ ਇਕ) 12 ਅੰਕ
5. ਭਾਗ ਅ-II ਵਿੱਚ ਲੇਖ ਰਚਨਾ ਸੰਬੰਧੀ ਪ੍ਰਸ਼ਨ (ਤਿੰਨ ਵਿੱਚੋਂ ਇੱਕ) 8 ਅੰਕ
6. ਭਾਗ ੲ ਵਿੱਚ 'ਜੀਵਨੀ' ਅਤੇ 'ਵਿਆਕਰਨ ਅ-I' ਵਾਲੇ ਭਾਗ ਵਿੱਚੋਂ ਸੰਖੇਪ ਉੱਤਰਾਂ ਵਾਲੇ 15 ਲਾਜ਼ਮੀ ਪ੍ਰਸ਼ਨ ਪੁੱਛੇ ਜਾਣਗੇ।ਹਰੇਕ ਪ੍ਰਸ਼ਨ 2 ਅੰਕਾਂ ਦਾ ਹੋਵੇਗਾ। 15x2=30 ਅੰਕ

ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ **30 ਅੰਕ**

ਕਲਾਸ ਹਾਜ਼ਰੀ, ਘਰੇਲੂ ਇਮਤਿਹਾਨ, ਅੰਦਰੂਨੀ ਪ੍ਰਯੋਗੀ ਅਭਿਆਸ

ਨੋਟ : ਪ੍ਰਯੋਗੀ ਅਭਿਆਸ ਲਈ ਅਧਿਆਪਕ ਵੱਲੋਂ ਪਾਠਕ੍ਰਮ ਨਾਲ ਸੰਬਧਿਤ ਕਰਵਾਏ ਗਏ ਕਾਰਜਾਂ 'ਤੇ ਆਧਾਰਿਤ ਵਿਦਿਆਰਥੀਆਂ ਦੁਆਰਾ ਇੱਕ ਫਾਈਲ ਤਿਆਰ ਕੀਤੀ ਜਾਵੇਗੀ ਜਿਸ ਦੇ ਅੰਕ ਨਿਸ਼ਚਿਤ ਹੋਣਗੇ ਅਤੇ ਇਹ ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ ਦਾ ਹਿੱਸਾ ਹੋਵੇਗੀ। ਇਸ ਲਈ ਇੱਕ ਪੀਰੀਅਡ ਪ੍ਰਤੀ ਹਫਤਾ ਲਗਾਇਆ ਜਾਵੇਗਾ

Approved

Chairperson
Board of Studies

ਸਹਾਇਕ ਪੁਸਤਕ ਸੂਚੀ

1 .ਹਰਕੀਰਤ ਸਿੰਘ (ਡਾ.),**ਰੂਪਾਂਤਰੀ ਵਿਆਕਰਣ**, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।

2. ਹਰਕੀਰਤ ਸਿੰਘ (ਡਾ.) ਤੇ ਗਿਆਨੀ ਲਾਲ ਸਿੰਘ, **ਕਾਲਜ ਪੰਜਾਬੀ ਵਿਆਕਰਣ ਤੇ ਲੇਖ-ਮਾਲਾ**, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।

3. ਜੋਗਿੰਦਰ ਸਿੰਘ ਪੁਆਰ ਅਤੇ ਹੋਰ, **ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦਾ ਵਿਆਕਰਨ** (ਭਾਗ-3), ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਅਕਾਦਮੀ, ਜਲੰਧਰ, 1994.

4. ਜਗਦੇਵ ਸਿੰਘ ਗਰੇਵਾਲ : **ਪੰਜਾਬੀ ਵਿਚ ਜੀਵਨੀ ਸਾਹਿਤ**, ਲਾਹੌਰ ਬੁੱਕ ਸ਼ਾਪ, ਲੁਧਿਆਣਾ।

5. ਦੁਨੀ ਚੰਦਰ, **ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦਾ ਵਿਆਕਰਨ**, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ, 1964.

6. ਧਰਮਪਾਲ ਸਿੰਗਲਾ (ਡਾ.) **: ਪੰਜਾਬੀ ਜੀਵਨੀ ਸਰੂਪ, ਸਿਧਾਂਤ ਤੇ ਵਿਕਾਸ,** ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ।

7. ਨਰਿੰਦਰ ਦੁਲੇ : **ਪੰਜਾਬੀ ਵਿਚ ਜੀਵਨੀ ਸਾਹਿਤ**, ਰਵੀ ਸਾਹਿਤ ਪ੍ਰਕਾਸ਼ਨ, ਅੰਮ੍ਰਿਤਸਰ।

8. ਬੂਟਾ ਸਿੰਘ ਬਰਾੜ, **ਪੰਜਾਬੀ ਵਿਆਕਰਨ: ਸਿਧਾਂਤ ਤੇ ਵਿਹਾਰ**,ਚੇਤਨਾ ਪ੍ਰਕਾਸ਼ਨ, ਲੁਧਿਆਣਾ, 2008.

Approved

Chairperson
Board of Studies

ਭਾਗ-ਤੀਜਾ (ਸਮੈਸਟਰ-ਪੰਜਵਾਂ)

BCA 504 A: ਪੰਜਾਬੀ ਲਾਜ਼ਮੀ (ਮੁੱਢਲਾ ਗਿਆਨ/Elementary Punjabi)

ਕ੍ਰੈਡਿਟ:4

ਕੁੱਲ ਅੰਕ:100 **ਵਿਸ਼ੇ ਵਿੱਚ ਪਾਸ ਅੰਕ: 35%**

ਲਿਖਤੀ ਪੀਖਿਆ: 70 ਅੰਕ **ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ: 30 ਅੰਕ**

ਲਿਖਤੀ ਪੀਖਿਆ ਦਾ ਸਮਾਂ: 3 ਘੰਟੇ **ਅਧਿਆਪਨ ਦੇ ਪੀਰੀਅਡ ਪਤੀ ਹਫ਼ਤਾ: 05**

ਨੋਟ : ਵਿਦਿਆਰਥੀਆਂ ਵੱਲੋਂ ਪਾਸ ਹੋਣ ਲਈ ਲਿਖਤੀ ਪੀਖਿਆ ਅਤੇ ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ ਦੇ ਅੰਕਾਂ ਵਿੱਚ ਅਲੱਗ-ਅਲੱਗ ਤੌਰ 'ਤੇ 35% ਅੰਕ ਲੈਣੇ ਜ਼ਰੂਰੀ ਹਨ।

Course Outcome:

1. ਬਾਹਰਲੇ ਰਾਜਾਂ ਦੇ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਸਬੰਧੀ ਜਾਣਕਾਰੀ ਹਾਸਿਲ ਕਰਦੇ ਹੋਏ ਇਸ ਦੇ ਇਤਿਹਾਸਕ ਪਰਿਪੇਖ, ਗੁਰੂਆਂ-ਪੀਰਾਂ, ਇਤਿਹਾਸਕ ਸਮਾਰਕਾਂ, ਪਿੰਡਾਂ, ਲੋਕਾਂ, ਲੋਕ ਸਾਹਿਤ ਅਤੇ ਲੋਕ ਖੇਡਾਂ ਆਦਿ ਰਾਹੀਂ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਨੂੰ ਸਮਝਣ ਅਤੇ ਇਸ ਵਿੱਚ ਇਕ ਜ਼ਿੰਮੇਵਾਰ ਨਾਗਰਿਕ ਵਜੋਂ ਵਿਚਰਨ ਦੇ ਯੋਗ ਹੋ ਜਾਂਦੇ ਹਨ।

2. ਵਿਹਾਰਕ ਪੰਜਾਬੀ ਵਿਆਕਰਨ ਦੇ ਗਿਆਨ ਰਾਹੀਂ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬ ਵਿੱਚ ਦਫ਼ਤਰੀ ਕੰਮ-ਕਾਜ ਕਰਨ ਦੇ ਯੋਗ ਹੋ ਜਾਂਦੇ ਹਨ।

3. ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦੇ ਉਚਾਰਨ ਸੰਬੰਧੀ ਮੁਹਾਰਤ ਹਾਸਲ ਕਰਦੇ ਹਨ ।

ਪਾਠ ਕਮ ਅਤੇ ਪਾਠ ਪੁਸਤਕ

ਭਾਗ - ੳ ਪਾਠ ਪੁਸਤਕ : ਮੁੱਢਲਾ ਪੰਜਾਬੀ ਗਿਆਨ- ਭਾਗ ਤੀਜਾ, ਮੁਖ ਸੰਪਾਦਕ ਪੋ. ਲਖਵੀਰ ਸਿੰਘ, ਸੰਪਾਦਕ ਡਾ. ਸੁਰਜੀਤ ਸਿੰਘ, ਪੋ. ਬਲਬੀਰ ਸਿੰਘ, ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ ਦੇ 1 ਤੋਂ 5 ਤੱਕ ਲੇਖ ।

ਭਾਗ - ਅ ਵਿਹਾਰਕ ਪੰਜਾਬੀ ਵਿਆਕਰਨ

(ਅ.1) ਪੈਰਾ ਪੜ ਕੇ ਪਸ਼ਨਾਂ ਦੇ ਉਤਰ ਦੇਣਾ।

(ਅ.2) ਰਿਪੋਰਟ ਰਾਈਟਿੰਗ

ਭਾਗ - ੲ ਬਹੁਤੇ ਸ਼ਬਦਾਂ ਦੀ ਥਾਂ ਇਕ ਸ਼ਬਦ ।(ਇਹਨਾਂ ਸ਼ਬਦਾਂ ਲਈ ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ ਪਟਿਆਲਾ ਦੁਆਰਾ ਪਕਾਸ਼ਿਤ ਕੀਤੀ ਗਈ ਉਪਰੋਕਤ ਲੇਖਾਂ ਵਾਲੀ ਪੁਸਤਕ ਦੇ ਪੰਨਾ ਨੰ:80 ਤੋਂ 86ਤੱਕ ਨੂੰ ਹੀ ਅਧਾਰ ਬਣਾਇਆ ਜਾਵੇ)

ਭਾਗ – ਸ ਪਾਠ ਕਮ ਦੇ ਭਾਗ ੳ ਉਤੇ ਅਧਾਰਿਤ ਸੰਖੇਪ ਉਤਰਾਂ ਵਾਲੇ ਪਸ਼ਨ।

ਅੰਕ ਵੰਡ ਅਤੇ ਪੇਪਰ ਸੈਟਰ ਲਈ ਹਦਾਇਤਾਂ

1. ਪਸ਼ਨਾਂ ਦਾ ਪੱਧਰ ਵਿਦਿਆਰਥੀਆਂ ਦੇ ਬੌਧਿਕ ਪੱਧਰ ਅਤੇ ਉਨਾਂ ਦੀ ਭਾਸ਼ਾ ਸਿੱਖਣ ਦੀ ਸੀਮਾ ਨੂੰ ਧਿਆਨ ਵਿਚ ਰੱਖ ਕੇ ਹੀ ਨਿਸ਼ਚਿਤ ਕੀਤਾ ਜਾਵੇ । ਉਨਾਂ ਨੂੰ ਸਰਲ ਤੇ ਸਪੱਸ਼ਟ ਪਸ਼ਨ ਹੀ ਪੁੱਛੇ ਜਾਣ।

2. ਲੋੜ ਅਨੁਸਾਰ ਵਿਦਿਆਰਥੀਆਂ ਨੂੰ ਛੋਟ ਜਾਂ ਚੋਣ ਦੇਣੀ ਲਾਜ਼ਮੀ ਹੈ।

3. ਪਸ਼ਨ ਸਾਰੇ ਭਾਗਾਂ ਵਿੱਚ ਪੁੱਛੇ ਜਾਣ।

4. ਲੇਖ ਦਾ ਸਾਰ (ਦੋ ਵਿੱਚ ਇੱਕ) 10 ਅੰਕ

5. ਲੇਖ ਦਾ ਵਿਸ਼ਾ ਵਸਤੂ (ਦੋ ਵਿੱਚ ਇੱਕ) 10 ਅੰਕ

6. ਪੈਰਾ ਪੜ ਕੇ ਪੁੱਛੇ ਗਏ ਪੰਜ ਪਸ਼ਨਾਂ ਦੇ ਉਤਰ 2×5 = 10 ਅੰਕ

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7. ਰਿਪੋਰਟ ਰਾਈਟਿੰਗ-ਕਾਲਜ ਜਾਂ ਵਿਭਾਗ ਨਾਲ ਸਬੰਧਤ ਗਤੀਵਿਧੀਆਂ/ ਸਮਾਗਮਾਂ ਨਾਲ ਸਬੰਧਤ(ਤਿੰਨ ਵਿਚ ਇਕ) 10 ਅੰਕ

8. ਬਹੁਤੇ ਸ਼ਬਦਾਂ ਦੀ ਥਾਂ ਇਕ ਸ਼ਬਦ (ਪੰਦਰਾਂ ਵਿਚ ਦਸ) 10 ਅੰਕ

ਨੋਟ: ਇਹ ਸ਼ਬਦ ਸਿਰਫ ਮੁੱਢਲਾ ਪੰਜਾਬੀ ਗਿਆਨ, ਭਾਗ ਤੀਜਾ ਦੇ ਪੰਨਾ ਨੰ: 80 ਤ 86 ਤੱਕ, ਵਿਚ ਹੀ ਪੁੱਛੇ ਜਾਣ ।

9. ਭਾਗ ਓ ਉਤੇ ਅਧਾਰਿਤ ਸੰਖੇਪ ਉਤਰਾਂ ਵਾਲੇ 12 ਪਸ਼ਨ ਪੁੱਛੇ ਜਾਣਗੇ। ਜਿਨ੍ਹਾਂ ਵਿਚੋਂ ਵਿਦਿਆਰਥੀ ਨੇ 10 ਪ੍ਰਸ਼ਨਾਂ ਦੇ ਉੱਤਰ ਦੇਣੇ ਹੋਣਗੇ। 10×2=20 ਅੰਕ

ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ 30 ਅੰਕ

ਕਲਾਸ ਹਾਜ਼ਰੀ, ਘਰੇਲੂ ਇਮਤਿਹਾਨ, ਅੰਦਰੂਨੀ ਪਯੋਗੀ ਅਭਿਆਸ

ਨੋਟ: - ਅਧਿਆਪਕ, ਕਲਾਸ ਵਿਚ ਵਿਦਿਆਰਥੀਆਂ ਨੂੰ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਨੂੰ ਸ਼ੁੱਧ ਰੂਪ ਵਿੱਚ ਬੋਲਣ, ਲਿਖਣ ਦੀ ਮੁਹਾਰਤ ਪੈਦਾ ਕਰਨ ਅਤੇ ਉਨਾਂ ਦੀ ਬਹੁਪੱਖੀ ਸਖ਼ਸ਼ੀਅਤ ਉਸਾਰੀ ਲਈ ਪੈਕਟੀਕਲ ਤੌਰ 'ਤੇ ਕੰਮ ਕਰਵਾਉਣਗੇ । ਬੌਧਿਕ ਵਿਕਾਸ ਲਈ ਸਮਾਜਿਕ ਵਰਤਾਰਿਆਂ ਨੂੰ ਡੂੰਘਾਈ ਵਿੱਚ ਸਮਝਣ ਦੀ ਚੇਤਨਾ ਪੈਦਾ ਕਰਨ ਦੇ ਨਾਲ-ਨਾਲ ਉਨਾਂ ਅੰਦਰ ਸਭਿਆਚਾਰਕ ਰੁਚੀਆਂ ਨੂੰ ਪਫੁੱਲਤ ਕਰਨ ਲਈ ਕਲਾਸ ਵਿਚ ਵਿਦਿਆਰਥੀਆਂ ਦੇ ਪਾਠ-ਕਮ ਨਾਲ ਸਬੰਧਤ ਗਤੀਵਿਧੀਆਂ ਨੂੰ ਵੀ ਪੈਕਟੀਕਲ ਦਾ ਹਿੱਸਾ ਬਣਾਉਣਗੇ। ਇਸ ਲਈ ਇਕ ਪੀਰੀਅਡ ਪਤੀ ਹਫ਼ਤਾ ਲਗਾਇਆ ਜਾਵੇਗਾ। ਵਿਦਿਆਰਥੀ ਪੈਕਟੀਕਲ ਫਾਇਲ ਵਾਸਤੇ ਦੋ ਅਸਾਇਨਮਟ ਤਿਆਰ ਕਰਨਗੇ ।ਪਹਿਲੀ ਅਸਾਇਨਮਟ ਵਿੱਚ ਵਿਦਿਆਰਥੀ ਕਾਲਜ ਦੇ ਕਿਸੇ ਇਕ ਸਮਾਗਮ/ਵਿਭਾਗੀ ਗਤੀਵਿਧੀ ਜਾਂ ਵਿਦਿਅਕ ਦੌਰੇ ਸੰਬੰਧੀ ਰਿਪੋਰਟ ਤਿਆਰ ਕਰਨਗੇ । ਦੂਸਰੀ ਅਸਾਇਨਮਟ ਵਿੱਚ ਵਿਦਿਆਰਥੀ ਪਾਠਕਮ ਨਾਲ ਸਬੰਧਤ ਲੇਖਾਂ ਵਿਚੋਂ ਕਿਸੇ ਇਕ ਲੇਖ ਨਾਲ ਸਬੰਧਤ ਤਸਵੀਰਾਂ ਲਗਾ ਕੇ ਉਸ ਦਾ ਸਾਰ/ਵਿਸ਼ਾ ਵਸਤੂ ਲਿਖਣਗੇ ।

ਸਹਾਇਕ ਪੁਸਤਕ ਸੂਚੀ

1. ਹਰਕੀਰਤ ਸਿੰਘ (ਡਾ.) ਤੇ ਗਿਆਨੀ ਲਾਲ ਸਿੰਘ, **ਕਾਲਜ ਪੰਜਾਬੀ ਵਿਆਕਰਨ ਤੇ ਲੇਖ-ਮਾਲਾ**, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ, ਜੀਤ ਸਿੰਘ ਜੋਸ਼ੀ, **ਸਭਿਆਚਾਰ ਅਤੇ ਲੋਕਧਾਰਾ**, ਵਾਰਿਸ ਸ਼ਾਹ ਫਾਊਂਡੇਸ਼ਨ, ਅੰਮਿਤਸਰ, 1991.
2. ਬਲਵੀਰ ਸਿੰਘ ਪੂਨੀ, **ਪੰਜਾਬੀ ਲੋਕਧਾਰਾ ਅਤੇ ਸਭਿਆਚਾਰ**, ਵਾਰਿਸ ਸ਼ਾਹ ਫਾਊਂਡੇਸ਼ਨ, ਅੰਮਿਤਸਰ, 1992.

ਨੋਟ : Only those Students who have not Studied Punjabi up to Matriculation can opt for Elementary Punjabi (Mudla Punjabi Gyan). Other Students Will Study Compulsory Punjabi.

Approved

Chairperson
Board of Studies

SMESTER V

BCA 505: ENGLISH COMMUNICATION SKILLS
4 Credits: 4H(L)

Time: 3 hours | **Max Marks: 100**
Periods per week: 5 | **Theory: 70 marks**
Pass Marks: 35% | **Internal Assessment: 30 marks**

Course Outcomes: After the successful completion of this course, students will be able to:

- Gain knowledge to analyze a written text critically
- Enhance practice in objective and subjective writing
- Increase vocabulary through the study of word parts, use of context clues, and practice with adictionary
- Build and enrich their vocabulary
- Enhance their reading, writing, listening and speaking skills in English
- Improve writing skills through essay writing and translation

SECTION A

Prescribed Text: *Ten One-Act Plays* by Department of English, G S S D G S Khalsa College, Patiala, 2020.
The following six plays are prescribed:

1) *The Boor:* Anton Chekov
2) *The New Hangman:* Laurence Housman
3) *The Bishop's Candlesticks:* Norman Mckinnel
4) *The Exchange:* Althea Thurston
5) *The Dear Departed:* William Stanley Houston
6) *Day of Atonement:* Margaret Wood**Testing:**

Q1. (a) One essay-type question with an internal alternative on central idea, character, incident or summary from the prescribed text in about 250 words. 12 marks
(b) Five short answer type questions to be attempted out of the given eight from the prescribed text in about 30 words each. 5X2=10 marks

SECTION-B

Prescribed Text: *The Written Word* by Vandana R. Singh, OUP, 2006.
Q2. Writing Skills:

(a) Transcoding: Prose to Dialogue
Testing: Transcode the given prose passage into 20 dialogues (10 dialogue exchanges)
10 marks

(b) Writing a bio-sketch of a renowned personality
Testing: The candidate shall write a bio-sketch in about 200 words taking help from the given hintsabout a famous personality.
10 marks

(c) Essay Writing
Testing: One essay to be attempted out of the given two. 10 marks

(d) Translation
Testing: The student will be asked to translate a passage of ten sentences from English toPunjabi/Hindi.
In lieu of Translation
(For Foreign/ Other State students who do not know Punjabi/Hindi)
Paraphrase of a stanza from any poem about 8 to 10 lines. 10 marks

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Q3.Vocabulary
Prescribed Text: *The Students' Companion* by Wilfred D. Best, Harper Collins Publishers, 2010

1. Abbreviations (pages 95-97)
   Testing: Attempt all 8 abbreviations

   4 marks

2. Words Frequently Misspelt (Pages 141-144)
   Testing: Attempt all 8 Misspelt words

   4 marks

INTERNAL ASSESSSMENT

30 MARKS

Internal Assessment will be given on the basis of attendance, MSTs and over-all performance in the class. There will also be an internal viva based on the following topics of practical relevance. The teacher should focus on enhancing the skills of the students in writing, speaking and reading. One period per week will be allotted for covering these topics.

Topics to be Covered:

1. Group Discussion on the given topic
2. Review of a book or a movie
3. Enacting a character from prescribed plays
4. Speaking a few lines about a famous personality

Note: A project file has to be made covering all the above topics

Books Recommended for Grammar and Composition:

- R. Singh, Vandana. *The Written Word*. Oxford University Press, 2006.
- Singh, Achhru and Dr. Dharminder Singh Ubha. *Personality Development and Soft Skills*.New Academic Publishing Co., 2011.
- Sehgal, M.K. Vandana Khetarpal. *Business Communication*. Excel Books, 2009.

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BCA 506: (A) MANAGEMENT INFORMATION SYSTEM

4 Credits: (3L+1T)

Time: 3hours **Maximum Marks: 100**
Pass Marks: 35% **Internal Assessment: 30 Marks**
Theory: 70 Marks

Course Objective

The objective of this course is to enlighten the importance of management and leadership in business information systems and help the students to improve their decision-making.

Course Learning Outcomes

After completion of this course students will be able to:

CO1: Apply the sound managerial concepts and principles in the development and operations of information system.

CO2: Describe and managing the digital firm evolution.

CO3: Analyze and synthesize business information and systems to facilitate evaluation of strategic alternatives.

CO4: Demonstrate the Applications of Information System in Functional Areas (Marketing and Financial).

INSTRUCTION FOR THE PAPER SETTER

The question paper will consist of three sections A, B and C**.** Section A and B will have four questions from the respective section of the syllabus carrying 10 marks for each question. Section C will consist of 10-15 short answer type questions carrying a total of 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES

The Candidates are required to attempt at-least two questions each from the section A and B. Section C is compulsory.

UNIT-I

Management Information system: Meaning and definition, Role of information system, Nature and scope of MIS.

Information and system concepts: Definition and types of information, Information quality, dimensions of information, value of information, general model of human as an information processor. System related concepts, elements of a system, and types of system.

Role and importance of Management: Introduction, levels and functions of management. Structure and classification of MIS, Components of MIS.

Framework for understanding MIS: Robert Anthony's hierarchy of management activity, Information requirements and levels of management.

UNIT-II

Decision making: Types of decisions, methods of choosing among alternatives, Role of MIS in decision making. Simon's model of decision making, Structured and unstructured decisions.

Development of MIS: Stages in the development of MIS, System development approaches: Waterfall model, Prototyping, Iterative enhancement model and Spiral model.

Applications of information systems in Functional areas: Marketing MIS, Financial MIS, Production MIS, Personnel MIS.

Decision Support Systems: Definition and characteristics, MIS versus DSS, Tools and Models for decision support.

Suggested Readings:

1. D.P. Goyal, “Management Information Systems: Managerial Perspectives”, Macmillan India Ltd.
2. Robert G. Murdick, Joel E. Ross, James R. Claggett, “Information Systems for Modern Management”, Prentice Hall of India Pvt. Ltd.
3. Gordon B. Davis, M. H. Olson, “Management Information Systems: Conceptual Foundations, Structure & Development”, McGraw Hill Book Co.
4. W.S. Jawadekar, “Management Information Systems”, Tata McGraw Hill Publishing Co.

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BCA506: (B) ARTIFICIAL INTELLIGENCE
4 CREDITS: (3L+1T)

Time: 3Hours **Maximum Marks: 100**
Pass Marks: 35% **Internal Assessment: 30 Marks Theory: 70 Marks**

Course Objective
To import basic proficiency in representing difficult real life problems in a state space representation so as to solve them using AI techniques like searching and game playing.

Course Learning Outcomes
After completion of this course students will be able to:
CO1: Understand the significance and domain of Artificial Intelligence and knowledge representation.
CO2: Analyze various search techniques for real life problems.
CO3: Acquire the knowledge of real world knowledge representation.
CO4: Design expert system.

INSTRUCTIONS FOR THE PAPER SETTER
The question paper will consist of three sections A, B & C. Section A and B will have four questions each from respective unit of the syllabus carrying 10 marks for each question. Section C will have 10 -15 short answer type questions, carrying total 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES
The candidates are required to attempt at least two questions each from sections A and B, and the entire section C.

UNIT-I
Introduction to Artificial Intelligence: Concept and definition importance of AI. History of AI, Turing Test. AI related fields. Nilsson's onion model.
Heuristic Search: Introduction basic concept of heuristic search, concept of heuristic knowledge types of heuristic research techniques.
Introduction to knowledge: Introduction, type of knowledge, knowledge representation, knowledge Storage, knowledge acquisition, Concept of knowledge base system.
Knowledge Representation Techniques: introduction, frames, semantic Network, conceptual graph, Conceptual Dependencies, Script, CYC.

UNIT-II
Prepositional Logic: Introduction, Syntax, Semantics, Inference methods in Prepositional Logic.
Predicate Logic: Introduction Syntax, Semantics of Predicate Logic, Clausal form, Resolution, Unification, and Inference Mechanisms.
Problem solving using AI: Introduction: Defining problem as state space, analyzing problem, finding the solution, Representation of AI problem. Example of AI problems: Tic-Tac-Toe, Water-jug problem, 8 - Puzzle problem, 8-queen problem, chess problem.
Expert Systems: Basic Components & Architecture of Expert Systems, ES- Shells, Game Playing-The Mini max Search Procedure. Planning-Overview, Components of Planning System.

Suggested Readings:

1. Introduction to Artificial Intelligence and Expert System by Dan W. Patterson, Published by Pearson Education, Inc
2. Introduction to Artificial Intelligence by E. Charnaik and D. McDermott, Addison-Wesley Publishing Company.
3. Expert System for Personal Computers by M Chandwick and J.A Hannah, Galgotia Publication Pvt. Ltd.
4. Artificial Intelligence: A Modern Approach by Russell and Peter Norvig, N.J: Prentice Hall

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SEMESTER-VI
BCA 601: WEB TECHNOLOGY-II
4 Credits: (3L+1T)

Time: 3hours **Maximum Marks: 100 Marks**
Pass Marks: 35% **Internal Assessment: 30**
Marks Theory: 70 Marks

Course Objective

This course aims to help students to acquire advanced web development applications using server side PHP programming and Database Connectivity with MySQL.

Course Learning Outcomes

After completion of this course students will be able to:

CO1: To design and develop the dynamic database driven web pages.

CO2: Helps to make the static website and turn it into the dynamic website run from a data base using PHP and MYSQL.

CO3: Learn different ways of connecting to MYSQL through PHP to create, retrieve and save the database.

CO4: Students will be able to design professional websites.

INSTRUCTIONS FOR PAPER-SETTER

The question paper will consist of three sections A, B and C. Section A and B will have four questions from the respective section of the syllabus carrying 10 marks for each question. Section C will consist of 10-15 short answer type questions carrying a total 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES

Candidates are required to attempt at least two questions each from sections A and B and the entire section C.

UNIT - I

Introduction: A Brief History of PHP, features, Installing and Configuring PHP.

PHP Basics- Keywords, Identifiers, Basic Syntax, Defining variable and constant, Data types, operators, Control Statements.

Arrays: Creating an Array, Indexed Array, Associative Array, Multidimensional Array, Functions related to arrays.

String Manipulations: Quoting String Constants, Printing Strings, Accessing Individual Characters, Cleaning Strings, Encoding and Escaping, Comparing Strings, Manipulating and Searching Strings, Regular Expressions.

UNIT - II

Functions: Defining a Function, Calling Functions and Scope of Variables: Global and Local Variables, Built-in functions.

Object Oriented PHP: Declaring a Class, Creating an Object, Accessing Properties and Methods, constructors, destructors, inheritance.

Handling HTML Form with PHP: Super global variables, Capturing Form Data, Generating File uploaded form, redirecting a form after submission, Session and Cookies handling.

Database Connectivity: Introduction to MySQL, data types, Connecting MySQL database, Performing basic database operations (Insert, Delete, Update, Select) using PHP.

Suggested Readings:

1. Steven M. Schafer, "HTML, CSS, Java Script, Perl, Python and PHP", Wiley India.
2. Robin Nixon, "Learning PHP, My SQL & Java Script (with jquery, css & HTML5)", Oreilly.
3. Antun Peicevic, "My SQL introduction", Geek University Press.
4. SheeriK. Cabral & Keith Murphy, "My SQL Administrator's Bible", Wiley Publishing Inc.

Approved

Chairperson
Board of Studies

BCA 601 (P): SOFTWARE LAB-X (BASED ON PAPER BCA 601)
2 Credits: 4H (P)

Maximum Marks: 50 **Maximum Time: 3Hours Internal Assessment: 15Marks** **Pass Marks: 35%**

Course Objective
This course aims to help students to acquire advanced web development applications using server side PHP programming and Database Connectivity with MYSQL.
Course Learning Outcomes
After completion of this course students will be able to:
CO1: To design and develop the dynamic database driven web pages.
CO2: Helps to make the static website and dynamic website using PHP and MYSQL.
CO3: Implement different ways of connecting to MYSQL through PHP for creating, retrieving, saving and connecting the database.
CO4: Students will be able to design professional websites.

This laboratory course will comprise as exercises to supplement what is learnt under paper BCA 6.1-Web Technology-II.
The breakup of marks for the practical will be as under: -

i. Internal Assessment	15Marks
ii. Viva Voce(External Evaluation)	15Marks
iii. Lab Record, Program Development and Execution (External Evaluation)	20Marks

Students are required to develop the following programs with internal documentation:
1. Write a PHP program to display "Hello World!!!" in Bold and Italic onscreen.
2. Write a PHP program to display today's date and current time.
3. Write a PHP program to display the Fibonacci series.
4. Write a PHP program to get values from HTML form using GET and POST method.
5. Write a PHP program to print table of 5 using for and while loop.
6. Write a PHP program to demonstrate the use of array.
7. Write a PHP program to prepare student mark sheet using switch and ladder if-else.
8. Write a PHP program to swap two variables with/without using third variable.
9. Write a PHP program to create, delete and copying file from PHP script.
10. Write a program to SET and GET session and cookie value.
11. Write a program to create database, table and fields.
12. Write a program to perform CRUD operation (create, read, update and delete) in my sql using PHP.
13. Write a PHP program to demonstrate the concept of joins in my sql.
14. Write a program to demonstrate the concept of my sql constraints.
15. Create a calculator web application that performs BODMAS (Brackets, Orders, Division, Multiplication, Addition and Subtraction), Square Root and Cube Root.
16. Create a Goods and Services Tax based billing system.

Approved

Chairperson
Board of Studies

BCA 602: E-TECHNOLOGIES
4 Credits: (3L+1T)

Time: 3hours **Maximum Marks: 100**
Pass Marks: 35% **Internal Assessment: 30**
Marks Theory: 70 Marks

Course Objective

The aim of this course is to improve the understanding of E-Commerce, data mining and business intelligence. It will also help the students to understand about online payment methods and security breaches.

Course Learning Outcomes

After completion of this course students will be able to:

CO1: Analyze the impact of E-commerce business models and latest technologies of IT.
CO2: Identity the process that should be followed in building the E-Commerce.
CO3: To manipulate graphics and the sound information, privacy and control of Electronic media.
CO4: Different E-commerce trading relationships between the Entrepreneur and the consumers.

INSTRUCTIONS FOR THE PAPER SETTER

The question paper will consist of three sections A, B and C. Section A and B will have four questions from
the respective section of the syllabus carrying 10 marks for each question. Section C will consist of 10-15 short answer type questions carrying a total 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES

Candidates are required to attempt two questions each from the sections A & B of the question paper and the entire section C.

UNIT–I

Introduction to E-Commerce: Definition, Elements and Applications of E-Commerce. Advantages and Disadvantages of E-Commerce, E-Commerce v/s Traditional Commerce, E-Commerce v/s E- Business, Architectural Frame work of E-Commerce.

Security and Web: World Wide Web as an Architecture, Hypertext Publishing, HTTP, HTTPS, SSL, Legal, Ethical, Public Policy issues related to E-Commerce.

E-commerce Models: Mercantile Process Model-Consumer's and Merchant's Perspective, Internet Advertising-Active or Push based advertising, Passive or Pull based advertising.

E-payments &Issues: EFT, EDI & its applications, Payment Modes-Debit Card, Credit Card, Smart Card, Digital Token & Wallet, E-Cash, Internet Banking-NEFT, RTGS & IMPS, Universal Payment Interface, Payment Gateways, Money transfer applications, Digital Frauds.

UNIT-II

Software Agents-Role and Characteristics of S/W Agents in E-Commerce, Technologies behind Software Agents, Components of Software Agent.

Data Warehousing- Need and Environment of Data warehouse, Architecture and Technical Framework of a data warehouse methodology, analysis, design, construction and administration. Federated and Hub and Spokes Architecture (Overview).

Data Mining- Concepts: Evolution of Data Mining, Data Mining Process Model. Data Mining Techniques: Classification, Regression.

Application of Data Mining: Financial, Retail, Biological, Intrusion Detection mining, mining scientific and business data.

Suggested Readings:

1. Elaine Lawrence, "Internet Commerce: Digital Models for Business", Wiley.
2. Ravi Kalakota, Andrew B. Whinston: Frontiers of Electronic Commerce, Addison Wesley.
3. Daniel Minoli, Emma Minoli, "Web Commerce Technology Handbook", McGraw Hill
4. Alex Berson, Stephen, J. Smith, "Data Warehousing, Data Mining and OLAP", McGraw Hi**l**.

Approved

Chairperson
Board of Studies

BCA 603: MINOR PROJECT
2 Credits: (4P)

Maximum Times: 3Hours. **Max Marks: 50Marks**
Min Pass Marks: 35% **Internal Assessment: 15 Marks**
Practical Sessions to be conducted: 40-50Hrs

Course Objective
The objective of this course is to test the programming abilities/coding skills of students and to improve the team building, communication and managements skills of students. It will also enable students to create very precise specifications of the IT solutions to be designed.

Course Learning Outcomes
After completion of this course students will be able to:
CO1: Identify the requirements for the real world problems and to understand the programming language concepts and basics of Software Development Life Cycle model for the implementation of the project.
CO2: Conduct a survey of several available literatures in the preferred field of study to enhance software/hardware skills.
CO3: Build the project by Planning, analyzing, coding, testing, debugging and implement it at the premises of end user.
CO4: Report and present the findings of the study conducted in the preferred domain by working in the teams.

This laboratory course will comprise of a project report, which is prepared on the basis of what is learnt in the previous semesters of BCA.

The breakup of marks for the practical will be as under

i. Lab Record(Internal Assessment	15Marks
ii. Viva Voce(External Evaluation)	15Marks
iii. Project Evaluation(External Evaluation)	20Marks

Note:
1. Maximum 3 students are allowed in a group for a project.
2. The Project must be done under the Supervision of concerned faculty and all phases of Software Engineering has to be discussed.
3. The month wise progress must be evaluated to maintain the lab record for the internal assessment.

Approved

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Board of Studies

BCA 604: SYSTEM SOFTWARE
4 Credits: (3L+1T)

Maximum Time: 3Hours **Maximum Marks: 100**
Pass Marks: 35% **Internal Assessment: 30 Marks**
Theory: 70 Marks

Course Objective
This course aims to provide better understanding of language processors and compilation process. It will also help the students to understand the basics of Interpreter, Text Editors and Debuggers.
Course Learning Outcomes
After completion of this course students will be able to:
CO1: Identify and illustrate the types and features of system software and application software.
CO2: Describe the importance of Language Processors.
CO3: Demonstrate the use of Compilation Process which helps in programming languages.
CO4: Identify and understand the working of key components of computer system, hardware, software and firmware.

INSTRUCTIONS FOR THE PAPER SETTER
The question paper will consist of three sections A, B & C. Section A and B will have four questions each
from respective unit of the syllabus carrying 10 marks for each question. Section C will have 10-15 short answer type questions, carrying total 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES
The candidates are required to attempt at least two questions each from sections A and B, and the entire section C.

UNIT-I
Introduction: Definition of system software, types of system software, features of system programming, system programming vs. application programming.
Language Processors: Introduction, Language processing activities, Fundamentals of Language Processing.
Assembler: Elements of Assembly Language Processing, A simple Assembly scheme, Pass structure of Assemblers, Design of a two-pass assembler. A brief overview of single pass assembly and problem of forward references.
Linkers and Loaders: Definition of linker and loader and its types.

UNIT-II
Compilers: Phases of Compilation Process, Lexical Phase, Syntax Phase, Interpretation Phase, Optimization Phase (Compile time evaluation, Elimination of common sub-expression, Dead code elimination, Frequency reduction, strength reduction – concepts only), Storage Assignment Phase, Code Generation Phase.
Interpreter: Types of interpreter, Advantages and Disadvantages.
Text Editors: Design of text editor, Editing process, Types of editors.
Debugger: Debugging functions and capabilities, breakpoint mechanism, types of debugger.

Suggested Readings:

1. Dhamdhere D. M. "Systems Programming and Operating system", Tata McGraw-Hill
2. John J. Donovan, "System programming", McGraw-Hill.
3. Aho and Ulman, "Principles of Compilers", Narosa Publishing House.
4. Aho, A.V., Ullman Sethi R., I.D.: Compilers: Principles, Techniques and Tools, Addison-Wesley Publishing Co.

Approved

Chairperson
Board of Studies

ਭਾਗ-ਤੀਜਾ, ਸਮੈਸਟਰ-ਛੇਵਾਂ
BCA 605 -ਪੰਜਾਬੀ ਲਾਜ਼ਮੀ

ਕ੍ਰੈਡਿਟ : 4 **ਅਧਿਆਪਨ ਦੇ ਪੀਰੀਅਡ ਪ੍ਰਤੀ ਹਫ਼ਤਾ : 05**
ਕੁੱਲ ਅੰਕ : 100 **ਵਿਸ਼ੇ ਵਿੱਚੋਂ ਪਾਸ ਅੰਕ : 35%**
ਲਿਖਤੀ ਪ੍ਰੀਖਿਆ : 70 ਅੰਕ **ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ : 30 ਅੰਕ**
ਨੋਟ : ਵਿਦਿਆਰਥੀਆਂ ਵੱਲੋਂ ਪਾਸ ਹੋਣ ਲਈ ਲਿਖਤੀ ਪ੍ਰੀਖਿਆ ਅਤੇ ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ ਦੇ ਅੰਕਾਂ ਵਿਚੋਂ ਅਲੱਗ-ਅਲੱਗ ਤੌਰ 'ਤੇ 35% ਅੰਕ ਲੈਣੇ ਜ਼ਰੂਰੀ ਹਨ।

Course Outcome

1. ਇਸ ਵਿਸ਼ੇ ਰਾਹੀ ਵਿਦਿਆਰਥੀ ਆਪਣੇ ਪਰੰਪਰਾਗਤ ਇਤਿਹਾਸਿਕ ਵਿਰਸੇ ਅਤੇ ਸੱਭਿਆਚਾਰ ਦੇ ਵਿਭਿੰਨ ਪੱਖਾਂ ਬਾਰੇ ਜਾਣਕਾਰੀ ਹਾਸਿਲ ਕਰਕੇ ਆਧੁਨਿਕ ਸਮ ਵਿੱਚ ਹੋ ਰਹੇ ਲੋਕਧਰਾਈ ਰੂਪਾਤਰਾਂ ਨੂੰ ਵੀ ਸਮਝਣ ਦੀ ਯੋਗ ਹੁੰਦੇ ਹਨ।

2.ਵਿਦਿਆਰਥੀ ਇਸ ਰਾਹੀ ਲੋਕ ਮਾਨਸਿਕਤਾ ਨੂੰ ਸਮਝਣ ਦੇ ਯੋਗ ਹੋ ਕੇ ਇੱਕ ਨਰੋਏ ਸਮਾਜ ਦੀ ਸਿਰਜਣਾ ਕਰਨ ਦੇ ਸਮਰੱਥ ਹੁੰਦੇ ਹਨ।

3.ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦੀ ਵਾਕ ਬਣਤਰ ਵਿੱਚ ਮੇਲ ਤੇ ਅਧਿਕਾਰ ਸਥਾਪਿਤ ਕਰਨ ਵਾਲੀਆਂ ਵਿਆਕਰਨਿਕ ਇਕਾਈਆਂ ਸੰਬੰਧੀ ਜਾਣਕਾਰੀ ਹਾਸਿਲ ਕਰਦਾ ਹੈ ਅਤੇ ਇਹ ਜਾਣਕਾਰੀ ਉਸ ਨੂੰ ਦੂਜੀਆਂ ਭਾਸ਼ਾਵਾਂ ਦੀ ਵਾਕ ਰਚਨਾ ਨੂੰ ਵੀ ਸਮਝਣ ਵਿੱਚ ਸਹਾਈ ਹੁੰਦੀ ਹੈ।

4.ਵਿਦਿਆਰਥੀ ਅੰਗਰੇਜ਼ੀ ਭਾਸ਼ਾ ਤ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਵਿੱਚ ਅਨੁਵਾਦ ਕਰਨ ਸਬੰਧੀ ਤਕਨੀਕੀ ਜਾਣਕਾਰੀ ਹਾਸਿਲ ਕਰਦਾ ਹੈ।

ਪਾਠ-ਕ੍ਰਮ ਅਤੇ ਪਾਠ-ਪੁਸਤਕ

ਭਾਗ-ੳ

ਲੋਕਧਾਰਾ ਦੀ ਭੂਮਿਕਾ (ਨਿਬੰਧ ਸੰਗ੍ਰਹਿ) - ਸੰਪਾਦਕ : ਡਾ. ਭੁਪਿੰਦਰ ਸਿੰਘ ਖਹਿਰਾ, ਡਾ. ਸੁਰਜੀਤ ਸਿੰਘ

ਭਾਗ-ਅ

ਅ.I. ਵਿਆਕਰਨ

1. ਕਾਰਕ ਅਤੇ ਕਾਰਕੀ ਸਬੰਧ
2. ਵਾਕਾਤਮਕ ਜੁਗਤਾਂ- ਮੇਲ ਅਤੇ ਅਧਿਕਾਰ

ਅ.II. ਸਰਲ ਅੰਗਰੇਜ਼ੀ ਪੈਰ੍ਹੇ ਦਾ ਪੰਜਾਬੀ ਵਿਚ ਅਨੁਵਾਦ

ਭਾਗ-ੲ

ਭਾਗ ੳ ਅਤੇ ਅ.I ਵਿਅਕਰਨ ਵਾਲੇ ਭਾਗ 'ਤੇ ਆਧਾਰਿਤ ਸੰਖੇਪ ਉੱਤਰਾਂ ਵਾਲੇ ਪੰਦਰਾਂ ਪ੍ਰਸ਼ਨ।

ਅੰਕ ਵੰਡ ਅਤੇ ਪੇਪਰ ਸੈਟਰ ਲਈ ਹਦਾਇਤਾਂ

1. ਪ੍ਰਸ਼ਨ ਪੱਤਰ ਨੂੰ ਉਪਰੋਕਤ ਨਿਰਧਾਰਿਤ ਪਾਠ ਕ੍ਰਮ ਦੇ ਅਨੁਸਾਰ ਤਿੰਨ ਭਾਗਾਂ (ੳ,ਅ ਅਤੇ ੲ) ਵਿੱਚ ਵੰਡਿਆ ਜਾਵੇ।
2. ਭਾਗ ੳ ਵਿੱਚੋਂ ਕਿਸੇ ਇਕ ਨਿਬੰਧ ਦਾ ਵਿਸ਼ਾ ਵਸਤੂ/ਸਾਰ। (ਦੋ ਵਿੱਚੋਂ ਇਕ) 12 ਅੰਕ
3. ਭਾਗ ੳ ਵਿੱਚੋਂ ਨਿਬੰਧਾਂ ਵਿਚ ਪੇਸ਼ ਵਿਚਾਰਾਂ ਸਬੰਧੀ ਪ੍ਰਸ਼ਨ। (ਚਾਰ ਵਿੱਚੋਂ ਦੋ) 4+4=08 ਅੰਕ
4. ਵਿਆਕਰਨ ਵਾਲੇ ਭਾਗ ਅ-I ਵਿਚੋਂ ਵਰਣਾਤਮਕ ਪ੍ਰਸ਼ਨ। (ਦੋ ਵਿੱਚੋਂ ਇਕ) 12 ਅੰਕ
5. ਭਾਗ ਅ-II ਵਿੱਚ ਸਰਲ ਅੰਗਰੇਜ਼ੀ ਪੈਰ੍ਹੇ ਦਾ ਪੰਜਾਬੀ ਵਿਚ ਅਨੁਵਾਦ। 08 ਅੰਕ
6. ਭਾਗ-ੲ ਵਿੱਚ 'ਲੋਕਧਾਰਾ ਦੀ ਭੂਮਿਕਾ' ਅਤੇ 'ਵਿਆਕਰਨ ਦੇ ਭਾਗ ਅ-I' ਵਿੱਚੋਂ ਸੰਖੇਪ ਉੱਤਰਾਂ ਵਾਲੇ 15 ਲਾਜ਼ਮੀ ਪ੍ਰਸ਼ਨ ਪੁੱਛੇ ਜਾਣਗੇ। ਹਰੇਕ ਪ੍ਰਸ਼ਨ 2 ਅੰਕਾਂ ਦਾ ਹੋਵੇਗਾ। 15x2=30 ਅੰਕ

ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ **30 ਅੰਕ**

ਕਲਾਸ ਹਾਜ਼ਰੀ, ਘਰੇਲੂ ਇਮਤਿਹਾਨ, ਅੰਦਰੂਨੀ ਪ੍ਰਯੋਗੀ ਅਭਿਆਸ।

ਨੋਟ: ਪ੍ਰਯੋਗੀ ਅਭਿਆਸ ਲਈ ਅਧਿਆਪਕ ਵੱਲੋਂ ਪਾਠਕ੍ਰਮ ਦੇ ਅਨੁਸਾਰ ਕਰਵਾਏ ਗਏ ਕਾਰਜਾਂ 'ਤੇ ਆਧਾਰਿਤ ਵਿਦਿਆਰਥੀਆਂ ਦੁਆਰਾ ਇੱਕ ਫਾਈਲ ਤਿਆਰ ਕੀਤੀ ਜਾਵੇਗੀ ਜਿਸ ਦੇ ਅੰਕ ਨਿਸ਼ਚਿਤ ਹੋਣਗੇ ਅਤੇ ਇਹ ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ ਦਾ ਹਿੱਸਾ ਹੋਵੇਗੀ। ਇਸ ਲਈ ਇੱਕ ਪੀਰਿਅਡ ਪ੍ਰਤੀ ਹਫ਼ਤਾ ਲਗਾਇਆ ਜਾਵੇਗਾ ।

Approved

Chairperson
Board of Studies

ਸਹਾਇਕ ਪੁਸਤਕ ਸੂਚੀ

1. ਹਰਕੀਰਤ ਸਿੰਘ (ਡਾ.), **ਰੂਪਾਂਤਰੀ ਵਿਆਕਰਣ**, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।
2. ਹਰਕੀਰਤ ਸਿੰਘ (ਡਾ.) ਤੇ ਗਿਆਨੀ ਲਾਲ ਸਿੰਘ, **ਕਾਲਜ ਪੰਜਾਬੀ ਵਿਆਕਰਣ ਤੇ ਲੇਖ-ਮਾਲਾ**, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।
3. ਕਰਨੈਲ ਸਿੰਘ ਥਿੰਦ, **ਪੰਜਾਬ ਦਾ ਲੋਕ ਵਿਰਸਾ**, ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ।
4. ਗੁਰਬਖ਼ਸ਼ ਸਿੰਘ ਫ਼ਰੈਂਕ, **ਸਭਿਆਚਾਰ ਅਤੇ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ**, ਵਾਰਿਸ ਸ਼ਾਹ ਫਾਉਂਡੇਸ਼ਨ, ਅੰਮ੍ਰਿਤਸਰ।
5. ਡਾ.ਜਸਵਿੰਦਰ ਸਿੰਘ, **ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ: ਪਛਾਣ-ਚਿੰਨ੍ਹ**, ਪੁਨੀਤ ਪ੍ਰਕਾਸ਼ਨ, ਪਟਿਆਲਾ।
6. ਜੋਗਿੰਦਰ ਸਿੰਘ ਪੁਆਰ ਅਤੇ ਹੋਰ, **ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦਾ ਵਿਆਕਰਨ** ਭਾਗ-3, ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਅਕਾਦਮੀ, ਜਲੰਧਰ,1994.
7. ਜੀਤ ਸਿੰਘ ਜੋਸ਼ੀ, **ਸਭਿਆਚਾਰ ਅਤੇ ਲੋਕਧਾਰਾ ਦੇ ਮੂਲ ਸਰੋਕਾਰ**, ਲਾਹੌਰ ਬੁੱਕ ਸ਼ਾਪ ਲੁਧਿਆਣਾ।
8. ਦੁਨੀ ਚੰਦਰ, **ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦਾ ਵਿਆਕਰਨ**, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ,1964.
9. ਬਲਬੀਰ ਸਿੰਘ ਪੂਨੀ, **ਪੰਜਾਬੀ ਲੋਕਧਾਰਾ ਅਤੇ ਸਭਿਆਚਾਰ**, ਵਾਰਿਸ ਸ਼ਾਹ ਫਾਉਂਡੇਸ਼ਨ,ਅੰਮ੍ਰਿਤਸਰ।
10. ਬਲਦੇਵ ਸਿੰਘ ਚੀਮਾ (ਡਾ.), **ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਵਿਗਿਆਨ: ਤਕਨੀਕੀ ਸ਼ਬਦਾਵਲੀ ਦਾ ਕੋਸ਼**, ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ।

Approved

Chairperson
Board of Studies

ਭਾਗ-ਤੀਜਾ (ਸਮੈਸਟਰ-ਛੇਵਾਂ)

BCA 605 A: ਪੰਜਾਬੀ ਲਾਜ਼ਮੀ (ਮੁੱਢਲਾ ਗਿਆਨ/Elementary Punjabi)

ਕ੍ਰੈਡਿਟ:4

ਕੁੱਲ ਅੰਕ : 100 | **ਵਿਸ਼ੇ ਵਿੱਚ ਪਾਸ ਅੰਕ : 35%**

ਲਿਖਤੀ ਪੀਖਿਆ : 70 ਅੰਕ | **ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ : 30 ਅੰਕ**

ਲਿਖਤੀ ਪੀਖਿਆ ਦਾ ਸਮਾਂ : 3 ਘੰਟੇ | **ਅਧਿਆਪਨ ਦੇ ਪੀਰੀਅਡ ਪਤੀ ਹਫ਼ਤਾ : 05**

ਨਟ : ਵਿਦਿਆਰਥੀਆਂ ਵੱਲ ਪਾਸ ਹੋਣ ਲਈ ਲਿਖਤੀ ਪੀਖਿਆ ਅਤੇ ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ ਦੇ ਅੰਕਾਂ ਵਿੱਚ ਅਲੱਗ-ਅਲੱਗ ਤੌਰ 'ਤੇ 35% ਅੰਕ ਲੈਣੇ ਜ਼ਰੂਰੀ ਹਨ।

Course Outcome:

1. ਬਾਹਰਲੇ ਰਾਜਾਂ ਦੇ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਸਬੰਧੀ ਜਾਣਕਾਰੀ ਹਾਸਿਲ ਕਰਦੇ ਹੋਏ ਇਸ ਦੇ ਇਤਿਹਾਸਕ ਪਰਿਪੇਖ, ਗੁਰੂਆਂ- ਪੀਰਾਂ, ਇਤਿਹਾਸਕ ਸਮਾਰਕਾਂ, ਪਿੰਡਾਂ, ਲੋਕਾਂ, ਲੋਕ ਸਾਹਿਤ ਅਤੇ ਲੋਕ ਖੇਡਾਂ ਆਦਿ ਰਾਹੀ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਨੂੰ ਸਮਝਣ ਅਤੇ ਇਸ ਵਿੱਚ ਇਕ ਜ਼ਿੰਮੇਵਾਰ ਨਾਗਰਿਕ ਵਜ ਵਿਚਰਨ ਦੇ ਯੋਗ ਹੋ ਜਾਂਦੇ ਹਨ।
2. ਵਿਹਾਰਕ ਪੰਜਾਬੀ ਵਿਆਕਰਨ ਦੇ ਗਿਆਨ ਰਾਹੀ ਵਿਦਿਆਰਥੀ ਪੰਜਾਬ ਵਿੱਚ ਦਫ਼ਤਰੀ ਕੰਮ-ਕਾਜ ਕਰਨ ਦੇ ਯੋਗ ਹੋ ਜਾਂਦੇ ਹਨ।
3. ਵਿਦਿਆਰਥੀ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦੇ ਉਚਾਰਨ ਸੰਬੰਧੀ ਮੁਹਾਰਤ ਹਾਸਲ ਕਰਦੇ ਹਨ ।

ਪਾਠਕਮ ਅਤੇ ਪਾਠ ਪੁਸਤਕ

ਭਾਗ - ੳ ਪਾਠ ਪੁਸਤਕ :ਮੁੱਢਲਾ ਪੰਜਾਬੀ ਗਿਆਨ- ਭਾਗ ਤੀਜਾ, ਮੁਖ ਸੰਪਾਦਕ ਪੋ. ਲਖਵੀਰ ਸਿੰਘ, ਸੰਪਾਦਕ ਡਾ. ਸੁਰਜੀਤ ਸਿੰਘ, ਪੋ. ਬਲਬੀਰ ਸਿੰਘ, ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ ਦੇ 6 ਤ 10 ਤੱਕ ਲੇਖ ।

ਭਾਗ- ਅ ਵਿਹਾਰਕ ਪੰਜਾਬੀ ਵਿਆਕਰਨ

ਅ.1 ਚਿੱਠੀ ਪੱਤਰ (ਨਿਜੀ ਅਤੇ ਦਫ਼ਤਰੀ ਪੱਤਰ)

ਅ.2 ਪੈਸੀ/ਸੰਖੇਪ ਰਚਨਾ

ਭਾਗ - ੲ ਮੁਹਾਵਰੇ (ਇਹਨਾਂ ਮੁਹਾਵਰਿਆਂ ਲਈ ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ ਪਟਿਆਲਾ ਦੁਆਰਾ ਪਕਾਸ਼ਿਤ ਕੀਤੀ ਗਈ ਉਪਰੋਕਤ ਲੇਖਾਂ ਵਾਲੀ ਪੁਸਤਕ ਦੇ ਪੰਨਾ ਨੰ:72 ਤ 79 ਤੱਕ ਨੂੰ ਹੀ ਅਧਾਰ ਬਣਾਇਆ ਜਾਵੇ)

ਭਾਗ - ਸ

ਭਾਗ ੳ ਵਾਲੇ ਲੇਖਾਂ ਵਿੱਚ ਸੰਖੇਪ ਪੁਤਰਾਂ ਵਾਲੇ ਪਸ਼ਨ

ਅੰਕ ਵੰਡ ਅਤੇ ਪੇਪਰ ਸੈਟਰ ਲਈ ਹਦਾਇਤਾਂ

1.ਪਸ਼ਨ ਦਾ ਪੱਧਰ ਵਿਦਿਆਰਥੀਆਂ ਦੇ ਬੌਧਿਕ ਪੱਧਰ ਅਤੇ ਉਨਾਂ ਦੀ ਭਾਸ਼ਾ ਸਿੱਖਣ ਦੀ ਸੀਮਾ ਨੂੰ ਧਿਆਨ ਵਿੱਚ ਰੱਖ ਕੇ ਹੀ ਨਿਸ਼ਚਿਤ ਕੀਤਾ ਜਾਵੇ ਉਨਾਂ ਨੂੰ ਸਰਲ ਅਤੇ ਸਪਸ਼ਟ ਪਸ਼ਨ ਹੀ ਪੁੱਛੇ ਜਾਣ ।

2. ਲੋੜ ਅਨੁਸਾਰ ਵਿਦਿਆਰਥੀਆਂ ਨੂੰ ਛੋਟ ਜਾਂ ਚੋਣ ਦੇਣੀ ਲਾਜ਼ਮੀ ਹੈ।

3. ਪਸ਼ਨ ਸਾਰੇ ਭਾਗਾਂ ਵਿੱਚ ਪੁੱਛੇ ਜਾਣ।

4. ਲੇਖ ਦਾ ਸਾਰ (ਦੋ ਵਿੱਚ ਇੱਕ) 10 ਅੰਕ

5. ਲੇਖ ਦਾ ਵਿਸ਼ਾ ਵਸਤੂ (ਦੋ ਵਿੱਚ ਇੱਕ) 10 ਅੰਕ

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6. ਚਿੱਠੀ ਪੱਤਰ (ਤਿੰਨ ਵਿੱਚ ਇੱਕ) 10 ਅੰਕ

7. ਪੈਸੀ 10 ਅੰਕ

8. ਮੁਹਾਵਰੇ (ਪੰਦਰਾ ਵਿਚ ਦਸ) 10 ਅੰਕ

ਨਟ:ਇਹ ਮੁਹਾਵਰੇ ਸਿਰਫ ਮੁੱਢਲਾ ਪੰਜਾਬੀ ਗਿਆਨ, ਭਾਗ ਤੀਜਾ ਦੇ ਪੰਨਾ ਨੰ: 72 ਤ 79 ਤੱਕ ਵਿਚ ਹੀ ਪੁੱਛੇ ਜਾਣ ।

9. ਭਾਗ ੳ ਤੇ ਅਧਾਰਿਤ ਸੰਖੇਪ ਉਤਰਾ ਵਾਲੇ 12 ਪਸ਼ਨ ਪੁੱਛੇ ਜਾਣਗੇ। ਜਿਨ੍ਹਾਂ ਵਿਚੋਂ ਵਿਦਿਆਰਥੀ ਨੇ 10 ਪ੍ਰਸ਼ਨਾਂ ਦੇ ਉੱਤਰ ਦੇਣੇ ਹੋਣਗੇ।

10×2=20 ਅੰਕ

ਅੰਦਰੂਨੀ ਮੁਲਾਂਕਣ 30 ਅੰਕ

ਕਲਾਸ ਹਾਜ਼ਰੀ, ਘਰੇਲੂ ਇਮਤਿਹਾਨ, ਅੰਦਰੂਨੀ ਪਯੋਗੀ ਅਭਿਆਸ

ਨਟ: - ਅਧਿਆਪਕ, ਕਲਾਸ ਵਿਚ ਵਿਦਿਆਰਥੀਆਂ ਦੀ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਨੂੰ ਸ਼ੁੱਧ ਰੂਪ ਵਿੱਚ ਬੋਲਣ ਅਤੇ ਲਿਖਣ ਦੀ ਮੁਹਾਰਤ ਪੈਦਾ ਕਰਨ ਲਈ ਅਤੇ ਉਨਾਂ ਦੀ ਬਹੁਪੱਖੀ ਸਖ਼ਸ਼ੀਅਤ ਉਸਾਰੀ ਲਈ ਪੈਕਟੀਕਲ ਤੌਰ 'ਤੇ ਕੰਮ ਕਰਵਾਉਣਗੇ। ਬੌਧਿਕ ਵਿਕਾਸ ਲਈ ਸਮਾਜਿਕ ਵਰਤਾਰਿਆਂ ਨੂੰ ਡੂੰਘਾਈ ਵਿੱਚ ਸਮਝਣ ਦੀ ਚੇਤਨਾ ਪੈਦਾ ਕਰਨ ਦੇ ਨਾਲ-ਨਾਲ ਉਨਾਂ ਅੰਦਰ ਸਭਿਆਚਾਰਕ ਰੁਚੀਆਂ ਨੂੰ ਪਫੁੱਲਤ ਕਰਨ ਲਈ ਕਲਾਸ ਵਿਚ ਵਿਦਿਆਰਥੀਆਂ ਦੇ ਪਾਠ-ਕਮ ਨਾਲ ਸਬੰਧਤ ਗਤੀਵਿਧੀਆਂ ਨੂੰ ਵੀ ਪੈਕਟੀਕਲ ਦਾ ਹਿੱਸਾ ਬਣਾਉਣਗੇ।ਇਸ ਲਈ ਇਕ ਪੀਰੀਅਡ ਪਤੀ ਹਫ਼ਤਾ ਲਗਾਇਆ ਜਾਵੇਗਾ। ਵਿਦਿਆਰਥੀ, ਪੈਕਟੀਕਲ ਫਾਇਲ ਵਿੱਚ ਪਾਠਕਮ ਨਾਲ ਸਬੰਧਤ ਲੇਖਾਂ ਵਿਚ ਦੋ ਲੇਖ ਚੁਣਨ ਉਪਰੰਤ ਉਹਨਾਂ ਵਿਚ ਕਿਸੇ ਇਕ ਲੇਖ ਨਾਲ ਸਬੰਧਤ ਤਸਵੀਰਾਂ ਲਗਾ ਕੇ ਉਸ ਲੇਖ ਦਾ ਸਾਰ ਲਿਖੇਗਾ ਅਤੇ ਦੂਜੇ ਲੇਖ ਦਾ ਤਸਵੀਰਾਂ ਲਗਾ ਕੇ ਵਿਸ਼ਾ ਵਸਤੂ ਲਿਖੇਗਾ ।

ਸਹਾਇਕ ਪੁਸਤਕ ਸੂਚੀ

1. ਹਰਕੀਰਤ ਸਿੰਘ (ਡਾ.) ਤੇ ਗਿਆਨੀ ਲਾਲ ਸਿੰਘ, **ਕਾਲਜ ਪੰਜਾਬੀ ਵਿਆਕਰਨ ਤੇ ਲੇਖ-ਮਾਲਾ**, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ.
2. ਜੀਤ ਸਿੰਘ ਜੋਸ਼ੀ, **ਸਭਿਆਚਾਰ ਅਤੇ ਲੋਕਧਾਰਾ**, ਵਾਰਿਸ ਸ਼ਾਹ ਫਾਊਂਡੇਸ਼ਨ, ਅੰਮਿਤਸਰ, 1991.
3. ਬਲਵੀਰ ਸਿੰਘ ਪੂਨੀ, **ਪੰਜਾਬੀ ਲੋਕਧਾਰਾ ਅਤੇ ਸਭਿਆਚਾਰ**, ਵਾਰਿਸ ਸ਼ਾਹ ਫਾਊਂਡੇਸ਼ਨ, ਅੰਮਿਤਸਰ, 1992.

ਨਟ : Only those Students who have not Studied Punjabi up to Matriculation can opt for Punjabi Compulsory (Mudla Gyan/Elementary Punjabi). Other Students Will Study Compulsory Punjabi.

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Chairperson
Board of Studies

BCA 606: ENGLISH COMMUNICATION SKILLS
4 Credits: 4H(L)

Time: 3 hours | **Max Marks: 100**
Periods per week: 5 | **Theory: 70 marks**
Pass Marks: 35% | **Internal Assessment: 30 marks**

Course Outcomes: After the successful completion of this course, students will be able to:

- Gain knowledge to analyze a written text critically
- Enhance practice in objective and subjective writing
- Increase vocabulary through the study of word parts, use of context clues, and practice with a dictionary
- Build and enrich their vocabulary
- Enhance their reading, writing, listening and speaking skills in English
- Improve writing skills through Email writing, Précis writing and Translation

SECTION-A

Prescribed Text: *Compendium of Selected Essays* edited by Department of English, GSSDGS Khalsa College, Patiala.The following Six essays are prescribed:-

1. Omens : Joseph Addison
2. On the Rule of the Road : A.G.Gardiner
3. Of the Open Sky : John Ruskin
4. A Houseful of Sparrows : Ranjit Lal
5. On Running After One's Hat : G.K.Chesterton
6. Bookshop Memories : George Orwell

Testing:

Q1. (a) One essay type question with an internal alternative on central idea, character, incident or summary from the prescribed text in about 250 words. 12 marks

(b) Five short answer type questions to be attempted out of the given eight from the prescribed text in about 30 words each. 5x2=10 marks

SECTION-B

Prescribed Text: *Communication Skills* by Sanjay Kumar & Pushp Lata, Oxford University Press, New Delhi,2014.

Q2. Writing Skills:

a) E-mail writing: Address, subject, content, complimentary close.
Testing: One E-mail to be attempted out of the given two.
10 marks

b) Précis Writing
Testing: Write a précis of a given passage comprising about 150 words.
8 marks

c) Acquiring Life Skills:
The following topics are to be studied:

- Interview Skills
- Presentation Skills
- Stress busters
- Group discussion
- Leadership skills
- Time management

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Testing: Student will be asked to attempt 3 topics out of the given four.

4x3= 12 marks

(d) Translation

Testing: The student will be asked to translate a passage of ten sentences from Punjabi/Hindi to English.

In lieu of Translation

(For Foreign students who do not know Punjabi/Hindi) Paraphrase of a stanza from a poem about 8 to 10 lines. 10 marks

Q3. Vocabulary

Prescribed Text: *The Students' Companion* by Wilfred D. Best, Harper Collins Publishers, 2010

1. Abbreviations (pages 98-100)

   Testing: Attempt all 8 abbreviations

   4 marks

2. Formation of nouns and adjectives (Pages123-125)

   Testing: Attempt all 8 words

   4 marks

INTERNAL ASSESSSMENT

30 MARKS

Internal Assessment will be given on the basis of attendance, MSTs and over-all performance in the class. Therewill also be an internal viva based on the following topics of practical relevance. The teacher should focus on enhancing the skills of the students in writing, speaking and reading. One period per week will be allotted for covering these topics.

Topics to be covered:

1. Hosting an event
2. Notice making
3. Telling about your qualities for a particular job
4. Giving presentation on a topic

Note: A project file has to be made covering all the above topics.

Books Recommended for Grammar and Composition:

- R. Singh Vandana. *The Written Word*. Oxford University Press, 2006.
- Singh, Achhru and Dr. Dharminder Singh Ubha. *Personality Development and Soft Skills*. NewAcademic Publishing Co., 2011.
- Sehgal, M.K. Vandana Khetarpal. *Business Communication*. Excel Books,2009

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BCA 607(A): CLOUD COMPUTING

4 CREDITS :(3L+1T)

Time: 3Hours **Maximum Marks: 100**
Pass Marks: 35% **Internal Assessment: 30**
Marks Theory: 70 Marks

Course Learning Outcomes
After completion of this course students will be able to:
CO1: Apply the fundamental concepts in cloud infrastructure to understand the single and multiple data centers.
CO2: Outline the role of hardware and software for third party.
CO3: Analyse the various cloud computing techniques to solve the problems on the cloud.
CO4: Meet the evolution, its applicability, benefits as well as current and future challenge.

INSTRUCTIONS FOR THE PAPER SETTER

The question paper will consist of three sections A, B & C. Section A and B will have four questions each from respective unit of the syllabus carrying 10 marks for each question. Section C will have 10 -15 short answer type questions, carrying total 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES

The candidates are required to attempt at least two questions each from sections A and B, and the entire section C.

UNIT-I

Introduction: Definition, Basic Concepts and Terminology of Cloud Computing, Characteristics of Cloud Computing, Limitations, Necessity of Cloud Computing and Benefits of Cloud Computing.
Historical Development: Principles of Parallel and Distributed Computing, Grid Computing, Utility Computing, Comparing grid with Cloud Computing System.
Cloud Computing Architecture: Architecture over view diagram, Layers, Frontend, Backend, Common cloud management platform.
Cloud Service Delivery: Cloud service, Cloud service model architectures, Infrastructure as a service (IaaS) architecture, Platform as a service (PaaS) architecture, Examples of PaaS software, Software as a service (SaaS) architecture, Examples of SaaS applications.

UNIT-II

Cloud Deployment Scenarios: Cloud deployment models, Public clouds, Hybrid clouds, Community, Virtual private clouds, Vertical and special purpose, Selection criteria for cloud deployment.
Virtualization: Characteristics of virtualization, Benefits of virtualization, Types of Virtualization in cloud computing, hypervisors.
Tenancy: Types of tenancy, Multi tenancy, Application Programming Interfaces (API).
Application Areas of Cloud Computing: Health, Education, Agriculture, Social Networking, Media and entertainment Industry, Scientific Research Areas.

Suggested Readings:

1. Raj Kumar Buyya, James Broberg, Andrezei M.Goscinski, “Cloud Computing: Principles and paradigms”, Wiley.
2. Michael Miller,” Cloud Computing”, Pearson Education India.
3. Barrie Sosinsky, “Cloud Computing Bible”, Wiley India.
4. Dr. Saurabh Kumar, "Cloud Computing: Insights into New–Era Infrastructure", Wiley India.

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Board of Studies

BCA 607(B): DIGITAL MARKETING

4 Credits: (3L+1T)

Time: 3 hours **Maximum Marks: 100**
Pass Marks: 35% **Internal Assessment: 30**
Marks Theory: 70 Marks

Course Objective

Understand how and why to use digital marketing for multiple goals within a larger marketing and/or media strategy.

Course Learning Outcomes

After completion of this course students will be able to:
CO1. Analyze the confluence of marketing, operations, and human resources in real-time delivery. **CO2**: Explain the role and importance of digital marketing in a rapidly changing business landscape **CO3:** Discuss the key elements of a digital marketing strategy
CO4: Demonstrate advanced practical skills in common digital marketing tools such as SEO, SEM, and Social media and Blogs.

INSTRUCTION FOR THE PAPER SETTER

The question paper will consist of three sections A, B and C**.** Section A and B will have four questions from the respective section of the syllabus carrying 10 marks for each question. Section C will consist of 10-15 short answer type questions carrying a total of 30 marks, which will cover the entire syllabus uniformly.

INSTRUCTIONS FOR THE CANDIDATES

The Candidates are required to attempt at-least two questions each from the section A and B. Section C is compulsory.

UNIT I

Digital Marketing: Introduction, Definition of digital marketing, Setting Digital Marketing, Evolution, fundamentals, its scopes, techniques , Challenges to digital marketing, Difference between traditional and Digital Marketing, future of Digital Marketing.
E-Marketing: Email, Newsletters, Digests, Dedicated emails, video marketing, sponsorship emails and transactional emails. Compare and contrast E-Marketing with Traditional Marketing. Cashless Payment, UPI, Smart card, Debit Card, Credit Card, NEFT, RTGS.
Online Advertisement: Introduction, Its advantages, paid versus organic advertisement, Pay per Click (PPC) Model, introduction to SMS marketing.
Social Media Marketing: Introduction, Types of social media, Whatsapp ,Facebook, setting up a Facebook Business page, YouTube, LinkedIn, Twitter, Instagram, content marketing and Blogging Social Media Marketing strategy: Setting up goals, finding out the target group, media user name, availability, monitoring the brands and competition.

UNIT II

Web Analytics: Basic concepts, importance of web analytics, types of web analytics. Utility of web analytic in digital marketing.
Search Engine Optimization (SEO): Basics of SEO, working of Search Engine, essentials of SEO & its guidelines .Role of keyword in SEO .On-Page optimization: Basics of website designing /development, HTML basics for SEO.
Of-Page optimization (Onsite): Introduction, Local Marketing of website depending on Locations, Promoting subsequent pages of the website.
Lead Generation: Understanding, need of lead generation, different lead generation techniques, importance of lead generation.

Suggested Readings:

1. Puneet Sinch Bhatia, Fundamentals of Digital Marketing First Edition, Publication Pearson.
2. Vandana Ahuja. Digital Marketing 1st Edition, Publication Oxford.
3. Ryan Deiss and Russ Hennesberry. Digital Marketing for Dummies, 2nd Edition, John Wiley & ons. Inc.
4. Eric Enge, Stephan Speneer. The Art of SEO, 3rd Edition, O'Reilly Media Inc.

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